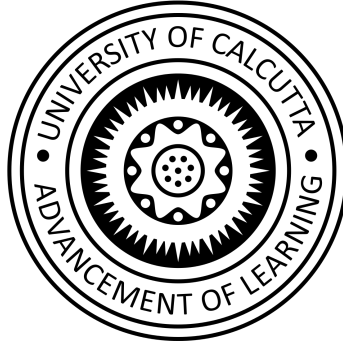




UNIVERSITY OF CALCUTTA

M.Sc. SEM-IV EXAMINATION, 2026
UNDER CBCS

SUBJECT: STATISTICS (STSA)
PAPER: 405

PROJECT TITLE

Statistical Modeling of Zoonotic Virus Pathogenicity
and Human Infection Risk Using Machine Learning

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July 2026

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Abstract

Zoonotic spillover—the transmission of pathogens from animal reservoirs to human populations—represents one of the most pressing challenges in contemporary epidemiology. Despite the growing global burden of vector-borne virus diseases, the biological and ecological determinants that distinguish human-infective viruses from those restricted to arthropod hosts remain incompletely understood. This study addresses that gap by developing a comprehensive statistical and machine learning framework for predicting human infection potential in a dataset of 896 virus records spanning 39 variables, sourced from the Mendeley Data Repository and compiled from WHO, NCBI, GenBank, and ICTV surveillance data.

The dataset integrates virus taxonomy, Baltimore genomic classification, arthropod vector biology, vertebrate reservoir host diversity, geographic distribution across six continents, and Köppen climatic classification. Exploratory analysis revealed that Arboviruses (85.7% of observations) exhibited significantly stronger zoonotic potential than Arthropod-Specific Viruses, with Group IV positive-sense ssRNA viruses dominating the dataset ($n = 544$). Inferential testing—including Chi-Square tests and Mann–Whitney U tests—confirmed statistically significant associations between human infectivity and virus group ($\chi^2 = 185.92$, $p < 0.001$), continental distribution ($\chi^2 = 21.65$, $p < 0.001$), cross-host diversity ($U = 3082.0$, $p < 0.001$), and total vector diversity ($U = 34646.5$, $p < 0.001$).

Following a systematic feature engineering process—incorporating multicollinearity filtering at $|r| \geq 0.75$, taxonomic lumping of rare virus families, and exclusion of data-leakage variables—a final set of 16 biologically meaningful predictors was retained. Five classification models were developed and evaluated: LASSO Logistic Regression (L1 regularization, $C = 0.4$; cross-validation accuracy ≈ 0.83), Ridge Logistic Regression (L2 regularization; cross-validation accuracy ≈ 0.87), Support Vector Machine (RBF and Linear kernels; best AUC = 0.924), Random Forest ($n_estimators = 200$; accuracy = 93.85%, AUC = 0.987), and XGBoost (accuracy = 94.97%, AUC = 0.99). While Ridge Logistic Regression demonstrated the strongest cross-validation performance, XGBoost achieved the highest test accuracy and produced the fewest false negatives (6 vs. 8 in Random Forest), making it the most suitable model for epidemiological surveillance applications where missed zoonotic detections carry significant public health costs.

Feature importance analysis consistently identified *Rodentia*, *Aves*, and *Chiroptera* as the most influential reservoir predictors, alongside Virus Group and Baltimore classification as key taxonomic determinants. These findings support the hypothesis that ecological host breadth and wildlife reservoir diversity are primary drivers of zoonotic emergence, and demonstrate the utility of machine learning in transitioning virus epidemiology from reactive outbreak response toward proactive predictive surveillance.

Chapter 1

Introduction

1.1 Introduction

Viruses are fundamental biological entities that exist at the edge of life, yet they represent some of the most significant challenges to global public health. While many viruses have evolved to persist within specific ecological niches, a subset possesses the unique ability to traverse species barriers. This phenomenon, known as zoonotic spillover, occurs when a pathogen is transmitted from vertebrate animal reservoirs to human populations. Historically, zoonotic diseases have been responsible for major global crises, ranging from the persistence of seasonal influenza to the emergence of highly pathogenic threats such as Ebola and the recent SARS-CoV-2 pandemic.

Traditionally, the identification of human-infective viruses has been a reactive process, where research is initiated only after an outbreak has reached a clinical threshold. However, with the vast diversity of the global virome, laboratory validation for every novel virus is logistically and economically unfeasible. There is a persistent knowledge gap regarding which biological signatures—taxonomic family, genetic replication mechanism (Baltimore group), or host range—provide the necessary flexibility for a virus to adapt to the human immune system.

Consequently, transitioning from reactive observation to proactive prediction is the next frontier in viral epidemiology.

This study addresses these challenges by employing a robust computational framework that integrates classical epidemiology with advanced machine learning.

Chapter 2

Objective of Study

The present study is directed toward the following specific objectives:

- To investigate how virus taxonomy, Baltimore genomic classification, arthropod vector taxonomy, geographical distribution, Köppen climatic classification, and vertebrate reservoir host diversity influence zoonotic virus pathogenicity and human infection risk.
- To perform exploratory data analysis and apply appropriate statistical tests — including Chi-Square tests of independence and Mann–Whitney U tests — to examine relationships between ecological predictors and human infectivity status across heterogeneous and non-normally distributed biological data.
- To build a smaller set of biologically meaningful predictor variables from the original 39-variable dataset through systematic removal of highly correlated variables, grouping of rare virus families, and exclusion of data-leakage variables, resulting in a final set of 16 ecologically interpretable variables.
- To apply regularization-based regression methods — specifically LASSO (L1) and Ridge (L2) Logistic Regression — for simultaneous variable selection, coefficient shrinkage, and identification of the most influential ecological determinants of human infectivity.
- To develop and evaluate four machine learning classification models — LASSO Logistic Regression, Ridge Logistic Regression, Random Forest, and XGBoost — alongside Support Vector Machine (SVM) with multiple kernel functions (RBF, Linear, and Polynomial), providing a comprehensive comparative framework for predicting zoonotic spillover potential.
- To assess and validate model performance using standard classification metrics including Accuracy, Precision, Recall, F1-score, ROC–AUC score, Confusion Matrix analysis, cross-validation accuracy, and Brier Score–based MSE and RMSE evaluation.
- To interpret predictor variable contributions through LASSO coefficient shrinkage, Ridge coefficient analysis, Random Forest Gini-based variable importance, and XGBoost gain-based variable importance — identifying the main ecological and taxonomic factors that determine human infection risk.

- To identify the best-performing classification model for epidemiological surveillance applications, with particular emphasis on minimizing false negative predictions of human-infecting viruses.

Chapter 3

Methodology

3.1 Dataset Description

The dataset used in this study was obtained from the Mendeley Data Repository, a recognized open-access scientific data platform. The records were compiled from multiple peer-reviewed virological studies, global epidemiological surveillance reports, and genomic repositories including the World Health Organization (WHO), National Center for Biotechnology Information (NCBI), GenBank, and the International Committee on Taxonomy of Viruses (ICTV) database.

The dataset consists of **896 observations** and **39 variables**, where each observation corresponds to a unique virus record associated with a specific host–vector–environment interaction. The dataset integrates information across multiple scientific domains, including virus taxonomy, reservoir host ecology, arthropod vector biology, geographic distribution, and climatic conditions. The variables are of mixed types: categorical, numerical (count-based), and binary.

The principal response variable is `homo_infected`, a binary indicator defined as:

$$\text{homo_infected} = \begin{cases} 1, & \text{if the virus infects humans} \\ 0, & \text{otherwise} \end{cases}$$

Among the 896 observations, 500 viruses are classified as human-infecting and 396 as non-human-infecting.

3.2 Exploratory Data Analysis and Inferential Testing

Prior to model development, we carried out a comprehensive exploratory data analysis (EDA) to understand the data distribution, ecological composition, and relationships between variables in the dataset.

We examined frequency distributions for all categorical variables including Virus Group, virus taxonomy (genus, family), vector taxonomy (genus, family, order, class), Baltimore classification group, continent, and Köppen climatic classification. Count-based variables such as reservoir host frequencies and vector diversity indices were examined using boxplots and grouped frequency distributions to assess skewness and the presence of outliers.

3.2.1 Inferential Statistical Tests

We applied two inferential testing procedures to evaluate associations between ecological predictors and human infectivity status (`homo_infected`).

Chi-Square Test of Independence

We applied the Chi-Square test to assess association between categorical variables and the binary response. For two categorical variables with observed frequencies O_{ij} and expected frequencies E_{ij} , the test statistic is:

$$\chi^2 = \sum_i \sum_j \frac{(O_{ij} - E_{ij})^2}{E_{ij}}$$

We applied this test to (i) Virus Group versus `homo_infected` ($\chi^2 = 185.92$, $p < 0.001$), and (ii) Continent versus `homo_infected` ($\chi^2 = 21.65$, $p < 0.001$).

Mann–Whitney U Test

We applied the Mann–Whitney U test to compare distributions of non-normally distributed continuous variables between the two infectivity groups. The test statistic is:

$$U = \min(U_1, U_2), \quad U_1 = n_1 n_2 + \frac{n_1(n_1 + 1)}{2} - R_1$$

where R_1 is the sum of ranks for group 1. We applied this test to (i) cross-host diversity versus `homo_infected` ($U = 3082.0$, $p < 0.001$), and (ii) total vector diversity versus `homo_infected` ($U = 34646.5$, $p < 0.001$).

3.3 Variable Construction and Selection

The original 39-variable dataset was reduced to a final set of 16 biologically meaningful predictor variables through a two-stage systematic variable selection process.

3.3.1 Stage 1: Multicollinearity Filtering and Ecological Screening

We computed a Pearson correlation matrix across all predictor variables. Variables with $|r| \geq 0.75$ were identified as highly correlated and selectively removed based on biological redundancy. Nested taxonomic variables at multiple hierarchical levels (e.g., `vi_G`, `vector_G`, `vector_O`, `vector_C`; and cumulative diversity indices `vector_G_T`, `vector_F_T`) were excluded to avoid multicollinearity in linear models.

Variables with extreme data sparsity — specifically, host group variables where over 95% of observations contained zero values (e.g., Reptilia, Anura, Pilosa, Lagomorpha at initial stages) — were evaluated for exclusion.

Two major variable construction steps were applied:

- **Taxonomic Grouping:** Virus families with occurrence frequency below 5% were grouped into a “Rare Family” category to ensure statistical robustness within each group.

- **Climatic Grouping:** Köppen climate classification codes were reduced to five primary groups (A, B, C, D, E), preserving the essential environmental signal while reducing degrees of freedom.

3.3.2 Stage 2: Refined Exclusion for Leakage and Redundancy

Four variables identified in the candidate set were excluded on the following grounds:

- **homo** and **Count:** Excluded to prevent data leakage. The variable **homo** is a direct numerical precursor of the binary target; its inclusion would artificially inflate model accuracy. **Count** reflects surveillance intensity focused on known pathogens, introducing surveillance bias.
- **cross_host:** Excluded to remove linear redundancy, as this aggregate index is mathematically derived from the individual host order variables retained in the model.
- **cross_vector_F:** Excluded due to near-perfect multicollinearity with **cross_vector_G** ($|r| \geq 0.75$). The genus-level diversity metric was retained as it provides finer resolution of vector transmission flexibility.

3.3.3 Final Variable Set

Following the two-stage process, a final set of 16 predictor variables was retained, as summarized in Table 3.1.

Table 3.1: Final Predictor Variables and Data Types

Category	Variable	Type
Target	homo_infected	Binary (1/0)
Virus Characteristics	Virus Group	Categorical
	baltimore	Categorical
Vector Information	vector_C	Categorical
Ecological Reservoirs	Aves, Carnivora, Rodentia	Numerical (count)
	Chiroptera, Primates	Numerical (count)
	Didelphimorphia, Artiodactyla	Numerical (count)
	Eulipotyphla, Lagomorpha	Numerical (count)
	Pilosa, Diprotodontia	Numerical (count)
Geographic Context	continent	Categorical
Environmental Context	climate_group	Categorical

3.4 Modeling Strategy

We structured the analytical pipeline in two distinct stages.

Stage 1 — Variable Selection and Coefficient Interpretation: LASSO (L1) and Ridge (L2) Logistic Regression models were applied using a **70:30 stratified train-test split**. These models were used mainly to identify biologically informative predictors through coefficient shrinkage and to assess the directional influence of ecological variables on human infectivity. They were not used as the primary predictive classifiers.

Stage 2 — Predictive Classification: Random Forest, XGBoost, and Support Vector Machine (SVM) classifiers were developed as the primary predictive models using an **80:20 stratified train-test split** to maximize training data availability for ensemble and kernel-based learning.

We applied stratified splitting throughout to preserve the proportional distribution of infected and non-infected virus observations across both subsets.

3.5 Preprocessing and Variable Scaling

We used different scaling strategies based on model requirements:

- **LASSO and Ridge Logistic Regression:** We applied `RobustScaler`, scaling variables by median and interquartile range (IQR) to reduce the influence of extreme outliers common in ecological count data:

$$X_{\text{scaled}} = \frac{X - \text{Median}(X)}{\text{IQR}(X)}, \quad \text{IQR}(X) = Q_3 - Q_1$$

- **Random Forest and XGBoost:** No variable scaling was required, as tree-based algorithms partition data through threshold splitting and are inherently insensitive to variable magnitude.
- **Support Vector Machine:** We applied `StandardScaler`, scaling each variable to zero mean and unit variance:

$$X_{\text{scaled}} = \frac{X - \mu}{\sigma}$$

SVM is sensitive to variable scale because its optimization is based on Euclidean distances.

We addressed class imbalance in all models using balanced class weighting, where minority-class observations receive proportionally greater weight during training.

3.6 Classification Models

3.6.1 LASSO Logistic Regression

LASSO (Least Absolute Shrinkage and Selection Operator) uses an L1 penalty on the logistic regression objective, simultaneously shrinking coefficients and carrying out variable selection by forcing some coefficients to exactly zero:

$$\min_{\beta} \{ \|Y - X\beta\|_2^2 + \lambda \|\beta\|_1 \}$$

The model was trained using the `liblinear` solver with regularization parameter $C = 0.4$ (where $C = 1/\lambda$).

3.6.2 Ridge Logistic Regression

Ridge regression applies an L2 penalty that shrinks all coefficients toward zero without eliminating any predictor, reducing multicollinearity and stabilizing coefficient estimation:

$$\min_{\beta} \{ \|Y - X\beta\|_2^2 + k\|\beta\|_2^2 \}$$

The ridge estimator is $\hat{\beta}^R = (Z'Z + kI_p)^{-1}Z'V$. The model was trained using the `liblinear` solver with $C = 0.4$.

3.6.3 Random Forest

Random Forest is an ensemble learning algorithm that constructs multiple decision trees on bootstrap samples of the training data, selecting a random subset of variables at each node split. Final predictions are determined by majority voting across all trees. Key hyperparameters used: `n_estimators` = 200, `min_samples_split` = 5, `min_samples_leaf` = 2, `class_weight` = balanced, `random_state` = 42.

3.6.4 Support Vector Machine

SVM constructs an optimal separating hyperplane in a high-dimensional variable space that maximizes the margin between classes. The soft-margin optimization problem is:

$$\min_{w,b,\xi} \frac{1}{2}\|w\|^2 + C \sum_{i=1}^n \xi_i \quad \text{subject to} \quad y_i(w^T x_i + b) \geq 1 - \xi_i, \quad \xi_i \geq 0$$

Three kernel functions were evaluated: RBF, Linear, and Polynomial (degree 3), each with $C = 1.0$. Hyperparameter tuning via `GridSearchCV` with 5-fold cross-validation was subsequently applied to the best-performing kernel.

3.6.5 XGBoost Classifier

XGBoost (Extreme Gradient Boosting) is a sequential ensemble method that iteratively adds decision trees to correct prediction errors of prior trees, minimizing a regularized objective function:

$$\text{Obj} = \sum_{i=1}^n l(y_i, \hat{y}_i) + \sum_{k=1}^K \Omega(f_k), \quad \Omega(f) = \gamma T + \frac{\lambda}{2} \sum_{j=1}^T w_j^2$$

Optimal leaf weights are derived analytically as $w_j^* = -G_j/(H_j + \lambda)$, where G_j and H_j are the first- and second-order gradient statistics. Class imbalance was addressed via the `scale_pos_weight` parameter, computed as $|y = 0|/|y = 1|$.

3.7 Performance Evaluation Metrics

All models were evaluated using the following metrics, computed on the independent held-out test set:

- **Accuracy:** $\frac{TP + TN}{TP + TN + FP + FN}$

- **Precision:** $\frac{TP}{TP + FP}$
- **Recall (Sensitivity):** $\frac{TP}{TP + FN}$
- **F1-score:** $2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$
- **ROC–AUC:** Area under the Receiver Operating Characteristic curve, measuring discriminatory capability across all classification thresholds.
- **Confusion Matrix:** Decomposition of predictions into True Positives (TP), True Negatives (TN), False Positives (FP), and False Negatives (FN).
- **Brier Score–based MSE and RMSE:**

$$\text{MSE} = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{p}_i)^2, \quad \text{RMSE} = \sqrt{\text{MSE}}$$

where $\hat{p}_i$ is the predicted probability for observation i .

Macro-average, weighted-average, and micro-average variants of precision, recall, and F1-score were also reported to assess performance stability across class-imbalanced conditions. Cross-validation accuracy curves across regularization parameters were examined for LASSO and Ridge models to assess generalization stability.

From an epidemiological surveillance perspective, minimization of false negatives (FN) was treated as a priority criterion, as missed zoonotic pathogens carry substantially greater public health cost than false alarms.

Chapter 4

Exploratory Data Analysis

4.1 Dataset Description

The dataset used in this study is a comprehensive collection of virological, ecological, environmental, and epidemiological information related to virus pathogenicity and zoonotic transmission. The main goal of the dataset is to identify the biological and ecological factors that influence the ability of viruses to infect humans.

The dataset mainly focuses on comparing *Arboviruses* and *Arthropod-Specific Viruses (ASVs)*. Arboviruses are transmitted through arthropod vectors such as mosquitoes and ticks and are capable of infecting vertebrate hosts, including humans. In contrast, Arthropod-Specific Viruses replicate only within arthropods and do not infect vertebrates. This distinction provides an ideal framework for investigating the determinants of zoonotic spillover.

The dataset integrates information from multiple scientific domains, including:

- Virus taxonomy
- Reservoir host ecology
- Vector biology
- Geographic distribution
- Climatic conditions
- Human infection status

Such multidimensional information enables the use of advanced statistical and machine learning approaches to predict virus pathogenicity and identify the most important ecological drivers of human infection.

4.2 Variables in the Dataset

The dataset consists of 896 observations and 39 variables, where each observation corresponds to a unique virus record associated with a specific host-vector-environment interaction.

The dataset contains a mixture of categorical, numerical, count-based, and binary variables.

4.3 Dependent Variable

The main response variable is:

4.3.1 homo_infected

A binary variable indicating whether a virus can infect humans.

$$\text{homo_infected} = \begin{cases} 1, & \text{if the virus infects humans} \\ 0, & \text{otherwise} \end{cases}$$

This variable serves as the target variable for predictive modeling.

4.4 Independent Variables

The predictor variables can be grouped into several biologically meaningful categories.

4.4.1 1. Virus Classification Variables

These variables represent the taxonomic identity and biological classification of viruses.

Variable	Description
Virus_Group	Arbovirus or Arthropod-Specific Virus
virus	Virus species name
vi_G	Virus genus
vi_F	Virus family
baltimore	Baltimore classification group

These variables are important because viruses belonging to certain families, such as *Flaviviridae* and *Togaviridae*, are historically associated with higher zoonotic potential.

4.4.2 2. Vector Taxonomy Variables

These variables describe the arthropod vectors associated with virus transmission.

Variable	Description
vector_G	Vector genus
vector_F	Vector family
vector_O	Vector order
vector_C	Vector class

Examples include mosquitoes (*Culicidae*) and ticks (*Ixodidae*), which are major carriers of zoonotic viruses.

4.4.3 3. Geographic and Environmental Variables

These variables describe the ecological distribution of viruses.

Variable	Description
country	Country of virus detection
climate	Köppen climatic classification
continent	Geographic continent

Environmental conditions strongly influence vector survival, virus persistence, and transmission dynamics.

4.4.4 4. Reservoir Host Variables

The dataset includes several variables representing virus detection frequencies in animal hosts.

Examples include:

- Aves
- Rodentia
- Chiroptera
- Primates
- Carnivora
- Artiodactyla
- Reptilia

These variables are biologically important because reservoir hosts such as bats (*Chiroptera*) and rodents (*Rodentia*) are major sources of emerging infectious diseases.

4.4.5 5. Host and Vector Diversity Variables

These variables quantify ecological adaptability.

Variable	Description
cross_host	Number of distinct host orders infected
cross_vector_G	Diversity of vector genera
cross_vector_F	Diversity of vector families
cross_vector_O	Diversity of vector orders
cross_vector_C	Diversity of vector classes

Higher diversity values may indicate greater spillover capability and broader ecological adaptability.

4.5 Source of the Dataset

The dataset was obtained from the *Mendeley Data Repository*, a recognized open-access scientific data platform. The records were compiled from multiple peer-reviewed virological studies, global epidemiological surveillance reports, and genomic repositories such as:

- World Health Organization (WHO)
- National Center for Biotechnology Information (NCBI)
- GenBank
- ICTV virus taxonomy database

The dataset represents years of accumulated global virus surveillance and therefore provides a scientifically reliable foundation for statistical modeling and machine learning analysis.

4.6 Nature of the Dataset

The dataset is characterized by:

- High dimensionality
- Mixed variable types
- Biological complexity
- Ecological heterogeneity
- Nonlinear interactions
- Potential multicollinearity among predictors

4.6.1 Human Infection Status

The variable `homo_infected` indicates whether a virus can infect humans or remains restricted to non-human hosts. This variable is epidemiologically important because it reflects the zoonotic and pathogenic potential of different viruses.

Viruses classified as *Infected* have been reported to infect humans, whereas *Not Infected* viruses have no confirmed evidence of human infection based on available epidemiological records.

The distribution of this variable provides insight into the proportion of viruses associated with human infection and public health risk. Among the total observations, 500 viruses are classified as *Infected*, while 396 belong to the *Not Infected* category. The higher frequency of infected viruses suggests that a large portion of the dataset consists of viruses with zoonotic or epidemic potential.

From an epidemiological perspective, human-infecting viruses are often associated with broader host diversity, vector involvement, and higher transmission potential. Many

circulate through complex ecological systems involving wildlife reservoirs, arthropod vectors, domestic animals, and humans.

In contrast, viruses categorized as *Not Infected* may remain limited to animal or arthropod hosts or may currently lack evidence of human pathogenicity. Overall, the distribution shows that the dataset is strongly enriched with viruses capable of infecting humans, highlighting its relevance for public health surveillance and predictive modeling.

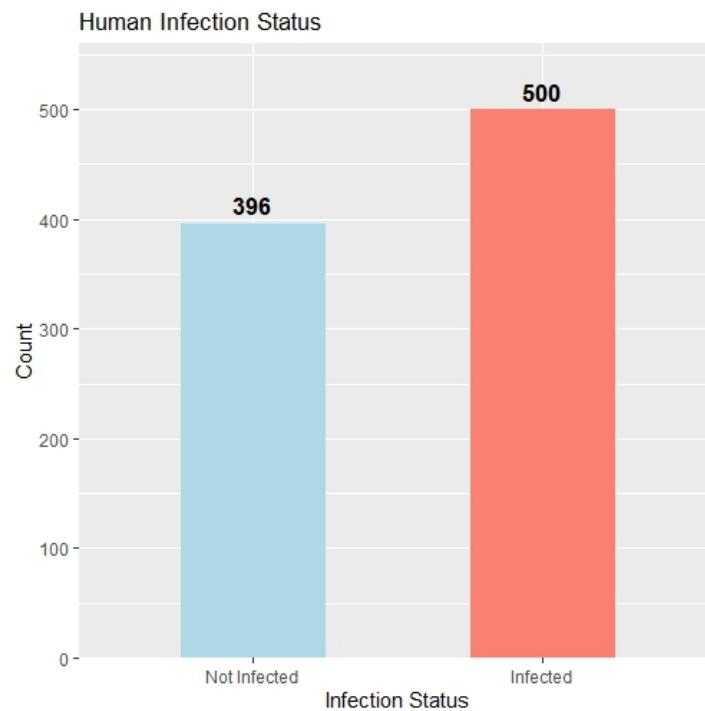


Figure 4.1: Distribution of Human Infection Status

4.6.2 Distribution of Virus_Group

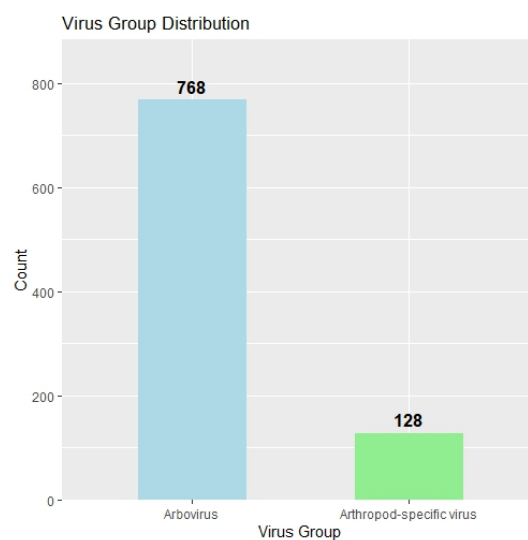


Figure 4.2: Distribution of Virus Groups

Figure 4.2 shows the distribution of viruses based on the variable `Virus_Group`. The dataset contains two major groups:

- **Arboviruses**
- **Arthropod-Specific Viruses**

Arboviruses are transmitted through arthropod vectors such as mosquitoes and ticks and are capable of infecting humans and other vertebrates. In contrast, Arthropod-Specific Viruses replicate mainly within arthropod hosts and generally do not infect humans.

The figure shows that the dataset is highly dominated by Arboviruses. Out of 896 total observations, 768 ($\approx 85.7\%$) belong to the Arbovirus group, while 128 ($\approx 14.3\%$) belong to the Arthropod-Specific Virus group.

This distribution suggests that the dataset mainly focuses on zoonotic and human-related viruses because of their greater public health importance. The imbalance between the two groups is also important for machine learning models, as it may introduce prediction bias toward the majority class.

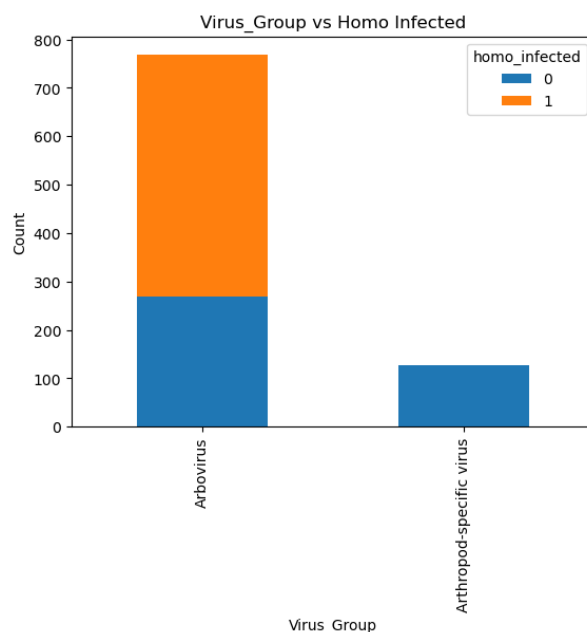


Figure 4.3: Relationship Between Virus Group and Human Infectivity

The figure effectively shows the strong association between `Virus_Group` and `homo_infected`.

The **Arbovirus** category contains a very high number of human-infecting viruses (1), represented by the large orange section in the stacked bar plot. In contrast, the **Arthropod-Specific Virus** category contains only non-human-infecting viruses (0), with no observed human infection cases.

This visual pattern shows that Arboviruses possess substantially greater zoonotic transmission potential compared to Arthropod-Specific Viruses, supporting the ecological role of vector-borne viruses in human infection and spillover dynamics.

4.6.3 Distribution of Virus Species

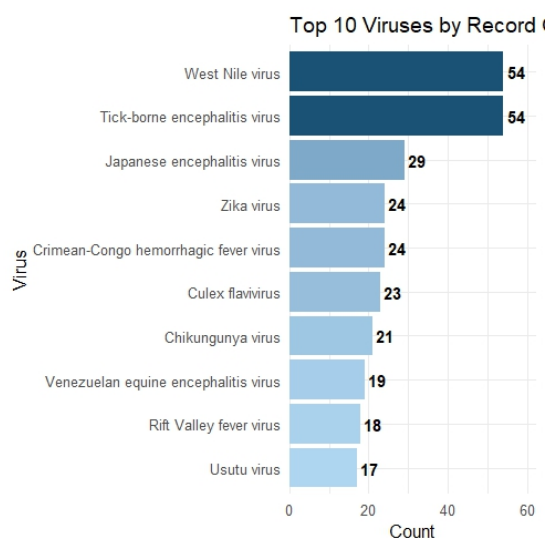


Figure 4.4: Top 10 Virus Species Based on Observation Frequency

Figure 4.4 shows the distribution of the top 10 virus species based on their frequency in the dataset.

The dataset contains 249 unique viruses, including both Arboviruses and Arthropod-Specific Viruses. Among them, West Nile virus and Tick-borne encephalitis virus have the highest number of observations, followed by Japanese encephalitis virus, Zika virus, and Crimean-Congo hemorrhagic fever virus.

Other important viruses in the dataset include Chikungunya virus, Rift Valley fever virus, Venezuelan equine encephalitis virus, Usutu virus, and Culex flavivirus. Many of these are mosquito- or tick-borne viruses associated with human infection, outbreaks, neurological diseases, and other public health concerns.

The higher frequency of these viruses reflects their epidemiological importance and the strong global surveillance focus on zoonotic and vector-borne diseases. This distribution highlights the relevance of the dataset for statistical analysis and predictive modeling of virus pathogenicity and zoonotic transmission.

4.6.4 Distribution of Virus Genera

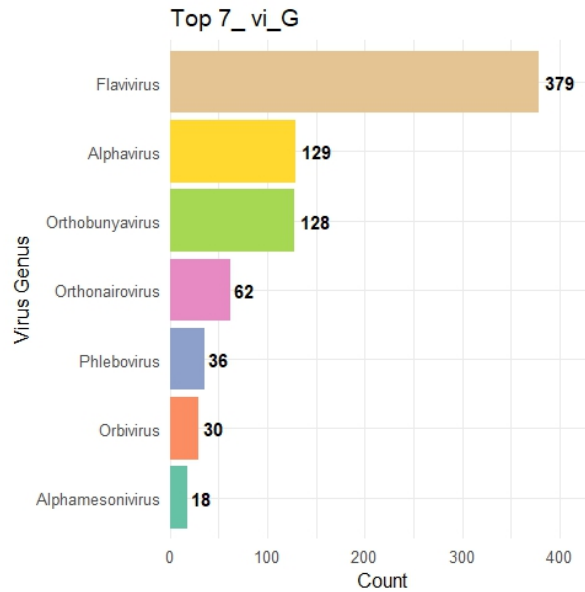


Figure 4.5: Distribution of Top 7 Virus Genera

Figure 4.5 shows the distribution of the top 7 virus genera based on their frequency in the dataset.

Among all genera, *Flavivirus* is the most dominant with 379 observations. This genus includes several important human pathogens such as Dengue virus, Zika virus, West Nile virus, and Japanese encephalitis virus. The second largest group is *Alphavirus* with 129 observations, followed by *Orthobunyavirus* with 128 observations.

Other important genera include *Orthonairovirus* (62 observations), *Phlebovirus* (36 observations), *Orbivirus* (30 observations), and *Alphamesonivirus* (18 observations).

The dominance of genera such as *Flavivirus*, *Alphavirus*, and *Orthobunyavirus* shows that the dataset mainly focuses on medically important vector-borne viruses associated with human infection and zoonotic transmission.

This taxonomic diversity is important for understanding host adaptability, vector specificity, transmission patterns, and the ecological factors influencing virus pathogenicity and zoonotic spillover.

4.6.5 Distribution of Virus Families

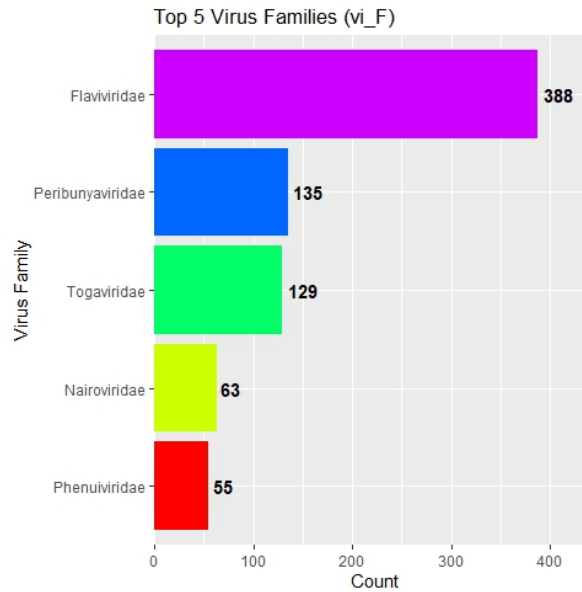


Figure 4.6: Distribution of Top Five Virus Families

Figure 4.6 presents the distribution of the top five virus families according to their frequency within the dataset.

The dataset shows substantial taxonomic diversity across multiple virus families associated with both Arboviruses and Arthropod-Specific Viruses. Figure 4.6 presents the top five virus families (*vi_F*) according to their frequency within the dataset.

Among all families, *Flaviviridae* represents the most dominant virus family with 388 recorded observations. This family includes several globally recognized arboviruses, including Dengue virus, Zika virus, West Nile virus, Yellow fever virus, and Japanese encephalitis virus. The dominance of *Flaviviridae* highlights its extensive geographic distribution and major contribution to vector-borne infectious diseases affecting human populations worldwide.

The family *Peribunyaviridae* is the second largest group with 135 observations, while *Togaviridae* follows closely with 129 observations. Viruses belonging to these families are widely associated with mosquito- and tick-borne transmission and are known to cause neurological disorders, febrile illnesses, and hemorrhagic manifestations in both humans and animals.

Other notable virus families identified in the dataset include *Nairoviridae* with 63 observations and *Phenuiviridae* with 55 observations. In addition, several less represented families such as *Rhabdoviridae*, *Reoviridae*, *Mesoniviridae*, *Asfarviridae*, and *Parvoviridae* further add to the biological and ecological heterogeneity of the dataset.

The inclusion of diverse virus families is epidemiologically significant because each family has unique genomic organization, replication strategies, host adaptability, vector specificity, and ecological transmission characteristics. Such taxonomic variability provides a broader framework for understanding the mechanisms underlying virus evolution, zoonotic spillover, and pathogenicity.

Furthermore, the dominance of medically important families such as *Flaviviridae*, *Peribunyaviridae*, and *Togaviridae* shows that the dataset is strongly enriched with

viruses of substantial public health importance. Consequently, comprehensive statistical and machine learning analyses of these virus families may provide valuable insights into transmission dynamics, host-vector interactions, ecological risk factors, and the determinants influencing virus emergence and human infectivity.

4.6.6 Distribution of Vector Genus

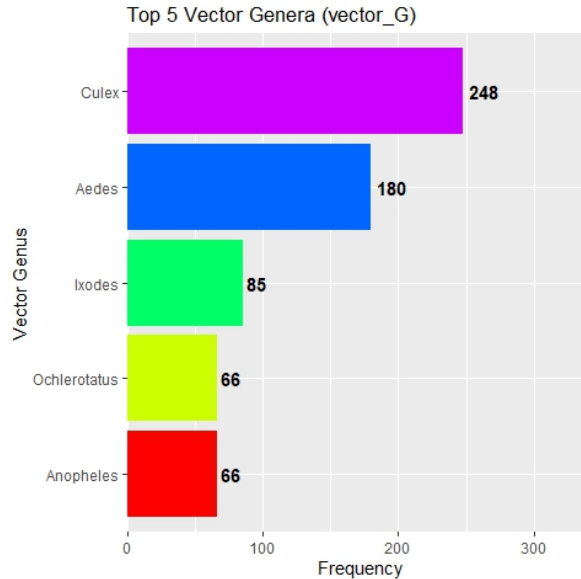


Figure 4.7: Distribution of Top Five Vector Genera

Figure 4.7 shows the distribution of vector genera in the dataset. A total of 38 vector genera are identified, among which the top 5 genera are presented in the figure, while the remaining genera are grouped under the category *Others*.

Among all vector genera, *Culex* is the most dominant with 248 observations, followed by *Aedes* with 180 observations and *Ixodes* with 85 observations. Other important genera include *Anopheles* and *Ochlerotatus*, each contributing 66 observations.

The dataset also contains other vector genera such as *Hyalomma*, *Ornithodoros*, *Culiseta*, *Culicoides*, and *Haemaphysalis*, which add to the ecological diversity of vector-borne pathogens.

The dominance of mosquito genera such as *Culex* and *Aedes* shows that mosquito-borne viruses form a major component of the dataset. Since vector diversity strongly influences transmission dynamics, host interaction, and zoonotic spillover risk, studying vector genera is important for understanding the epidemiology of emerging infectious diseases.

4.6.7 Distribution of Vector Families

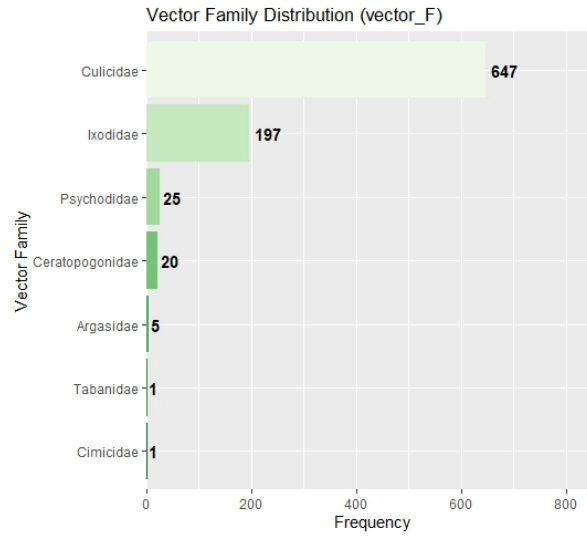


Figure 4.8: Distribution of Vector Families

Figure 4.8 shows the distribution of vector families associated with virus transmission in the dataset. A total of seven vector families are identified, including mosquitoes, ticks, sandflies, biting midges, bed bugs, and horse flies.

Among all families, *Culicidae* is the most dominant with 647 observations. This family includes mosquito genera such as *Aedes*, *Culex*, and *Anopheles*, which are major vectors of important arboviruses including Dengue virus, Zika virus, West Nile virus, and Japanese encephalitis virus.

The second largest family is *Ixodidae* with 197 observations, representing hard ticks associated with viruses such as Tick-borne encephalitis virus and Crimean-Congo hemorrhagic fever virus.

Other vector families include *Psychodidae* (25 observations), *Ceratopogonidae* (20 observations), *Argasidae* (5 observations), *Cimicidae* (1 observation), and *Tabanidae* (1 observation).

The dominance of *Culicidae* and *Ixodidae* shows that mosquito-borne and tick-borne viruses form the major component of the dataset. Vector family diversity is important for understanding transmission dynamics, ecological adaptation, and zoonotic spillover risk.

4.6.8 Distribution of Vector Orders

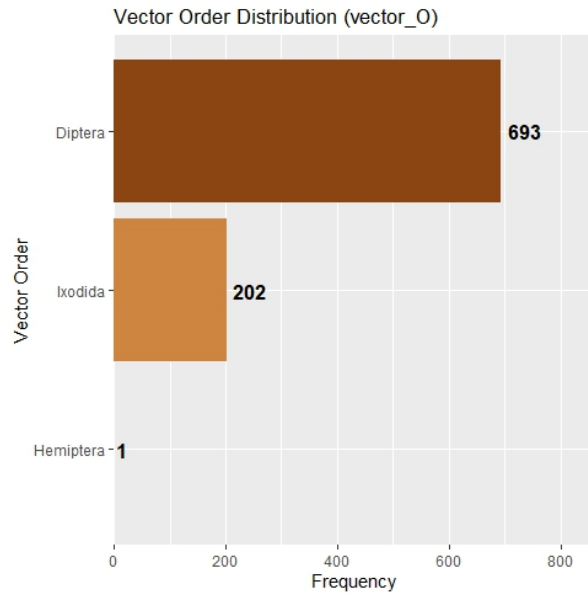


Figure 4.9: Distribution of Vector Orders

Figure 4.9 shows the distribution of vector orders in the dataset. Three vector orders are identified: *Diptera*, *Ixodida*, and *Hemiptera*.

Among all vector orders, *Diptera* is the most dominant with 693 observations. This order includes mosquitoes, sandflies, and biting midges such as *Aedes*, *Culex*, and *Anopheles*, which are major vectors of important arboviruses including Dengue virus, Zika virus, West Nile virus, and Japanese encephalitis virus.

The second largest order is *Ixodida* with 202 observations. This order includes hard and soft ticks associated with viruses such as Tick-borne encephalitis virus and Crimean-Congo hemorrhagic fever virus.

The order *Hemiptera* is represented by only one observation in the dataset.

The dominance of *Diptera* and *Ixodida* shows that mosquito-borne and tick-borne viruses form the major component of the dataset. Vector order diversity is important for understanding transmission pathways, ecological adaptation, and zoonotic spillover risk.

4.6.9 Distribution of Vector Classes

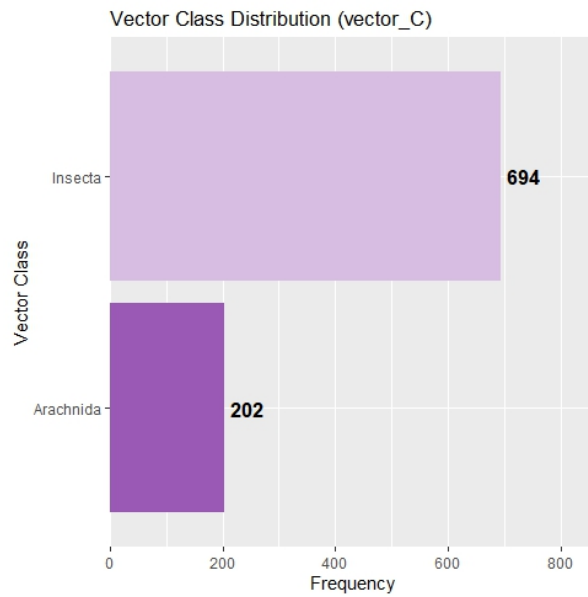


Figure 4.10: Distribution of Vector Classes

Figure 4.10 shows the distribution of vector classes in the dataset. Two major vector classes are identified: *Insecta* and *Arachnida*.

Among them, *Insecta* is the most dominant class with 694 observations. This class includes mosquitoes, sandflies, and other insects that play a major role in transmitting viruses to humans and animals.

In contrast, *Arachnida* contains 202 observations and mainly includes ticks and mites associated with several zoonotic and tick-borne viruses.

The higher prevalence of *Insecta* shows that insect-borne viruses form the major component of the dataset. Overall, the distribution highlights the ecological diversity of arthropod vectors and their importance in virus transmission dynamics and zoonotic spread.

The relationship between vector class (`vector_C`) and human infectivity status (`homo_infected`) is shown in Figure 4.11. The stacked bar plot compares the frequency of human-infecting and non-human-infecting viruses across different vector classes.

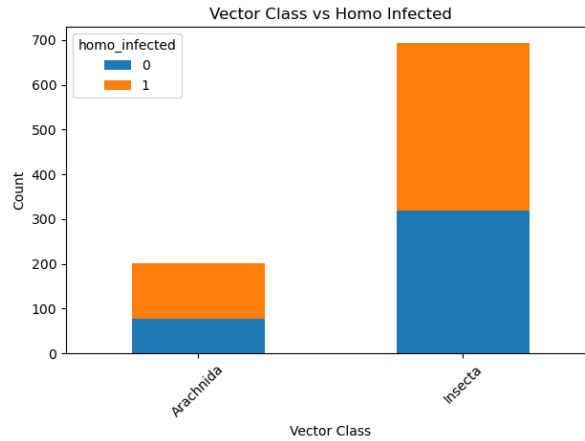


Figure 4.11: Relationship Between Vector Class and Human Infectivity

The figure shows that the **Insecta** vector class contains the highest number of viruses associated with human infection. A large proportion of viruses transmitted by insect vectors were classified as human-infecting, showing the major role of insect-mediated transmission in zoonotic spillover dynamics. In contrast, the **Arachnida** class also showed human-infecting viruses, but at a lower frequency.

This pattern suggests that insect vectors, particularly mosquitoes and related arthropods, play a major role in transmitting zoonotic viruses to humans. The dominance of the **Insecta** category highlights the epidemiological importance of vector-borne viral diseases in global public health.

4.6.10 Distribution of Baltimore Classification Groups

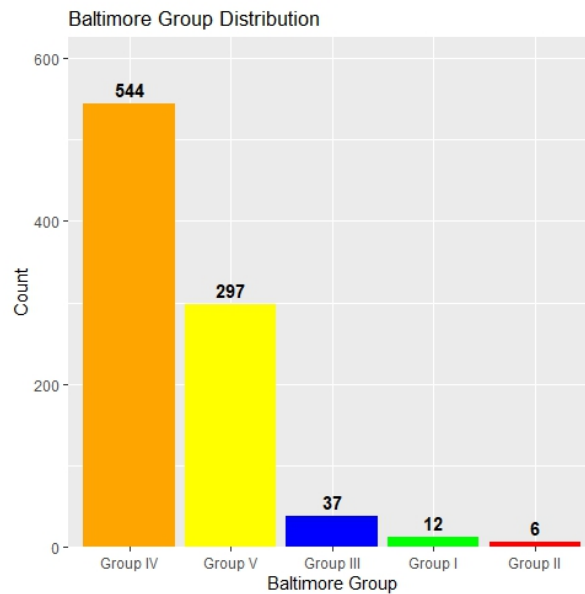


Figure 4.12: Distribution of Baltimore Classification Groups

Figure 4.12 presents the distribution of Baltimore classification groups observed in the dataset.

The Baltimore classification system groups viruses according to the type of nucleic acid in the virus genome and the way the virus produces messenger RNA (mRNA) for protein synthesis. Since every virus must make mRNA to replicate inside a host cell, this classification helps explain virus replication efficiency, mutation rate, transmission ability, and pathogenicity in humans.

The dataset contains a total of five Baltimore groups: Group I, Group II, Group III, Group IV, and Group V. The distribution of these groups shows considerable variation in their frequencies and biological importance.

Group IV viruses are the most dominant category in the dataset with 544 observations. These viruses contain positive-sense single-stranded RNA (+ssRNA), where the virus RNA directly acts as mRNA inside the host cell. As a result, protein synthesis and virus replication occur very rapidly. Group IV viruses also have high mutation rates and efficient transmission ability, making them highly adaptable human pathogens. Important examples include Dengue virus, Coronavirus, Hepatitis A virus, and Zika virus. Their rapid replication and widespread transmission explain their strong association with human infection and epidemic outbreaks.

Group V viruses represent the second largest category with 297 observations. These viruses contain negative-sense single-stranded RNA (-ssRNA). Unlike Group IV, their RNA cannot directly function as mRNA and must first be converted into complementary RNA before protein synthesis. Despite this additional step, Group V viruses remain highly infectious and pathogenic. Examples include Rabies virus and several influenza-related RNA viruses.

Group III viruses account for 37 observations in the dataset. These viruses have double-stranded RNA (dsRNA) genomes. Since host cells cannot directly translate dsRNA into proteins, the viruses must carry specialized enzymes to make mRNA. This added complexity generally results in a narrower infection range and lower transmission efficiency compared to Groups IV and V.

Group I viruses contain double-stranded DNA (dsDNA) and comprise 12 observations. These viruses can directly synthesize mRNA from DNA using host cell machinery. DNA viruses are usually more genetically stable and mutate more slowly than RNA viruses. As a result, their transmission patterns are often less explosive. Examples include Herpes viruses and Adenoviruses.

Group II viruses are the least represented category with only 6 observations. These viruses contain single-stranded DNA (ssDNA). Before replication, the ssDNA must first convert into double-stranded DNA and then produce mRNA, adding an extra replication step that slows virus multiplication. Parvoviruses are important examples of this group.

Overall, the distribution clearly shows that RNA viruses, especially Groups IV and V, dominate the dataset. Their rapid replication, high mutation rates, and enhanced adaptability make them better at infecting humans and spreading across populations. Based on biological behavior and pathogenic potential, the approximate order of human pathogenic strength observed in this study can be interpreted as:

$$\text{Group IV} > \text{Group V} > \text{Group I} > \text{Group III} > \text{Group II}$$

The dominance of Group IV and Group V viruses highlights the major epidemiological importance of RNA viruses in arthropod-borne and zoonotic viral diseases.

The relationship between Baltimore virus classification (`baltimore`) and human infectivity status (`homo_infected`) is shown in Figure 4.13. The stacked bar plot compares the

frequency of human-infecting and non-human-infecting viruses across different Baltimore groups.

The figure shows that Group IV viruses contain the highest proportion of human-infecting viruses among all Baltimore classifications. A large number of viruses belonging to this group were associated with human infection, showing the important role of positive-sense single-stranded RNA viruses in zoonotic spillover and virus transmission. Group V viruses also showed a considerable number of human-infecting viruses, although at a lower frequency compared to Group IV.

In contrast, Groups I, II, and III contained relatively few viruses and showed limited human infection cases. This uneven distribution suggests that virus genomic structure and replication mechanisms may strongly influence zoonotic potential and host adaptability.

The dominance of Group IV viruses highlights the epidemiological importance of RNA viruses in emerging infectious diseases and global public health.

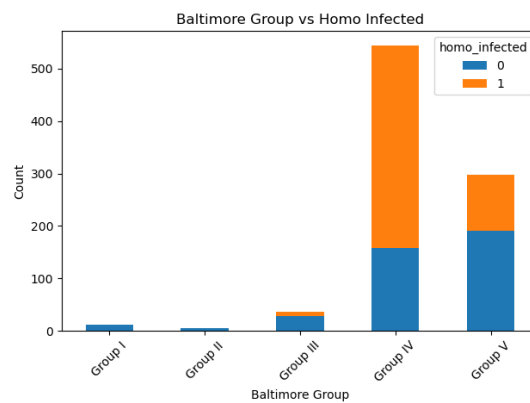


Figure 4.13: Relationship Between Baltimore Group and Human Infectivity

4.6.11 Distribution of Countries

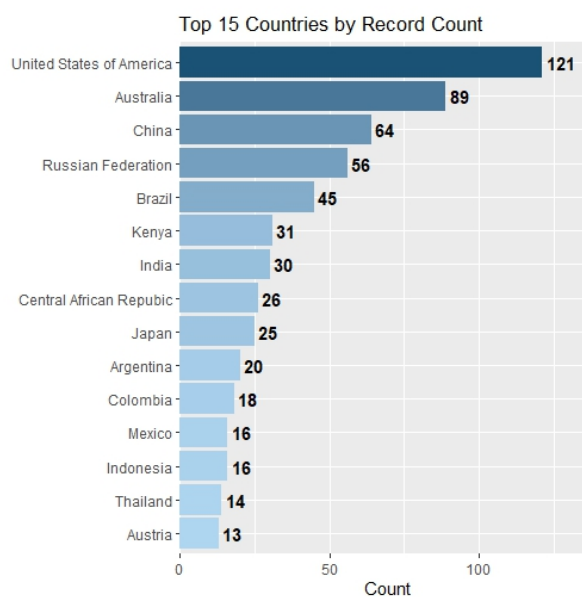


Figure 4.14: Country-wise Distribution of Virus Records

Figure 4.14 presents the geographical distribution of virus-related records across different countries included in the dataset.

The dataset contains virus-related records from a wide range of countries across multiple continents, reflecting the global distribution of arboviruses and arthropod-specific viruses. The geographical diversity observed in the dataset highlights the widespread occurrence of vector-borne viral diseases under different climatic and ecological conditions.

Among all countries, the United States of America contributes the highest number of records with 121 observations, followed by Australia with 89 observations, China with 64 observations, and the Russian Federation with 56 observations. Brazil contributes 45 observations, while Kenya and India contribute 31 and 30 observations respectively. These countries are recognized for substantial virus surveillance activities and ecological conditions favorable for vector survival.

Several tropical and subtropical countries also show notable representation, including the Central African Republic with 26 observations, Japan with 25 observations, Argentina with 20 observations, Colombia with 18 observations, and both Indonesia and Mexico with 16 observations each. Thailand and Austria contribute 14 and 13 observations respectively, while Trinidad and Tobago also contains 13 observations.

Additional countries such as Italy and South Africa each contribute 12 observations, whereas Germany, Canada, and the Republic of Korea contribute between 10 and 11 observations. Countries from Europe, Asia, Africa, and South America collectively add moderate and smaller frequencies, showing that virus circulation is globally distributed rather than region-specific.

The variation in country-wise frequencies may depend on several biological and environmental factors, including climate, vector abundance, biodiversity, population density, urbanization, international travel, and efficiency of public health surveillance systems. Tropical regions generally show higher vulnerability to arboviral transmission because warm and humid conditions support the breeding and survival of arthropod vectors such as mosquitoes and ticks.

The dataset also includes countries with lower observation frequencies, ranging from 1 to 9 records, including Yemen, Venezuela, Spain, Sri Lanka, Pakistan, Norway, Cuba, Cambodia, and several others. Although these countries add fewer observations, their inclusion shows the broad ecological reach and international importance of arthropod-associated viral diseases.

4.6.12 Distribution of Climate Types

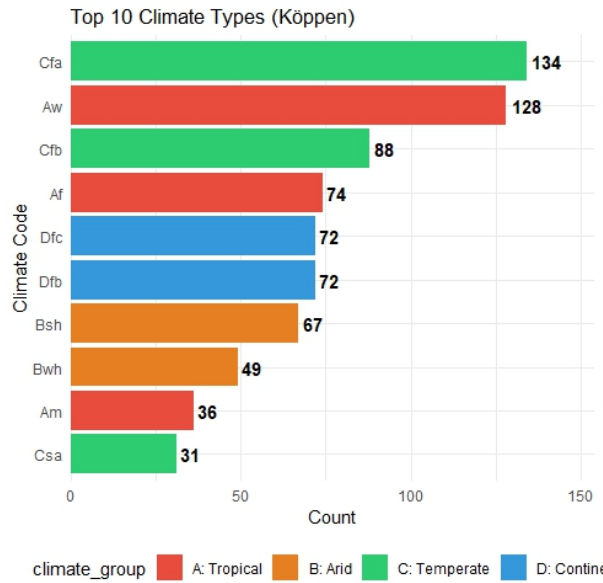


Figure 4.15: Distribution of Top Climate Types

Figure 4.15 presents the distribution of major climate types based on the Köppen climate classification system.

The dataset contains multiple climate types classified according to the Köppen climate classification system, which groups regions based on temperature, rainfall patterns, and seasonal variations. Climate plays a crucial role in the survival, breeding, and transmission efficiency of arthropod vectors such as mosquitoes and ticks. Therefore, the climatic distribution of the dataset provides important insight into the ecological conditions associated with virus transmission and human infection.

Among all climate categories, the most dominant climate type is Cfa (Humid Subtropical Climate) with 134 observations. This climate has hot summers, mild winters, and enough rainfall throughout the year, creating highly favorable conditions for mosquito breeding and vector survival. The second most frequent climate type is Aw (Tropical Savanna Climate) with 128 observations, followed by Cfb (Temperate Oceanic Climate) with 88 observations and Af (Tropical Rainforest Climate) with 74 observations.

Continental climate categories Dfb and Dfc each contribute 72 observations, showing that colder temperate regions also support some vector-borne viral activities. Arid and semi-arid climates such as Bsh (Hot Semi-Arid Climate) and Bwh (Hot Desert Climate) contribute 67 and 49 observations respectively, suggesting that some vectors can adapt even under relatively dry environmental conditions.

Additional climate categories include Am (Tropical Monsoon Climate) with 36 observations and Csa (Mediterranean Hot Summer Climate) with 31 observations. Other moderate-frequency climate types include Bsk with 27 observations, Cwa with 26 observations, Dfa with 22 observations, and Cwb with 18 observations. Several less frequent climates such as Bwk, Dwa, Dwb, Csb, As, Dfd, Dsb, Dwc, Dsa, and Et contribute between 1 and 9 observations.

For broader ecological interpretation, these Köppen climate types can be grouped into five major climate groups: Temperate, Tropical, Continental, Arid, and Polar climates.

These broader groups provide a simplified understanding of the environmental conditions influencing vector ecology and virus transmission dynamics.

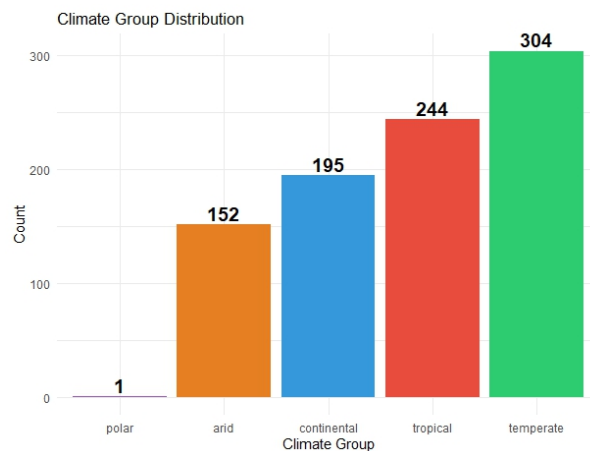


Figure 4.16: Distribution of Major Climate Groups

Figure 4.16 shows the distribution of broader climate groups within the dataset.

Among these climate groups, the Temperate climate group is the most dominant with 304 observations. Temperate regions generally have moderate temperatures and seasonal rainfall patterns, creating suitable conditions for the survival and persistence of arthropod vectors. The Tropical climate group contributes 244 observations and represents the second most dominant category. Tropical climates have high temperatures, high humidity, and abundant rainfall, all of which strongly support mosquito breeding and rapid virus replication.

The Continental climate group accounts for 195 observations. Although these regions have colder winters, seasonal warming allows the transmission of several vector-borne viruses. The Arid climate group contributes 152 observations, showing that some vectors can survive and adapt even in dry and semi-desert environments. In contrast, the Polar climate group is the least represented category with only 1 observation, reflecting the unfavorable environmental conditions for vector survival in extremely cold regions.

The dominance of tropical and humid temperate climates in the dataset shows that warm temperatures, high humidity, and seasonal rainfall strongly influence the transmission dynamics of arboviruses and arthropod-specific viruses. Such climatic conditions enhance vector reproduction, prolong vector lifespan, and increase the probability of virus transmission between hosts.

Tropical climates including Af, Am, and Aw are especially important for mosquito-borne diseases because high moisture and temperature speed up virus replication within vectors. Similarly, humid subtropical climates such as Cfa support continuous vector activity throughout much of the year, increasing the risk of endemic and epidemic outbreaks.

The relationship between climatic classification (`climate`) and human infectivity status (`homo_infected`) is shown in Figure 4.17. The stacked bar plot compares the frequency of human-infecting and non-human-infecting viruses across different climatic regions.

The figure shows that temperate and tropical climates contain the highest number of human-infecting viruses. A large proportion of viruses from these climatic regions were

associated with human infection, showing that environmental and ecological conditions may strongly influence zoonotic spillover dynamics and virus transmission patterns.

Continental and arid climates also showed moderate frequencies of human-infecting viruses, whereas polar climates showed very limited virus observations and almost no human infection cases. This uneven distribution suggests that climatic conditions may affect vector abundance, host diversity, and virus survival, thereby influencing the probability of zoonotic transmission.

The dominance of human-infecting viruses in tropical and temperate regions highlights the epidemiological importance of climate in shaping the global distribution of emerging infectious diseases.

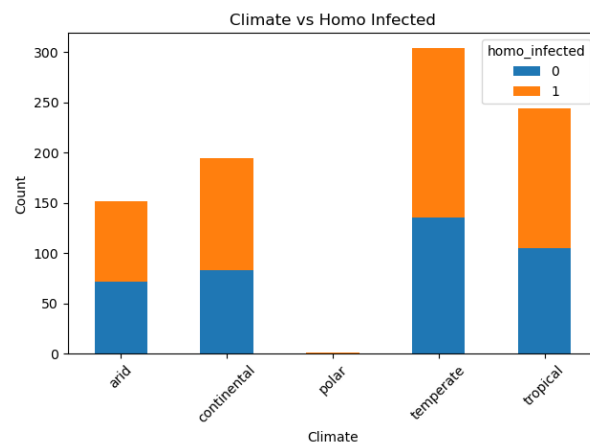


Figure 4.17: Relationship Between Climate and Human Infectivity

4.6.13 Distribution of Continents

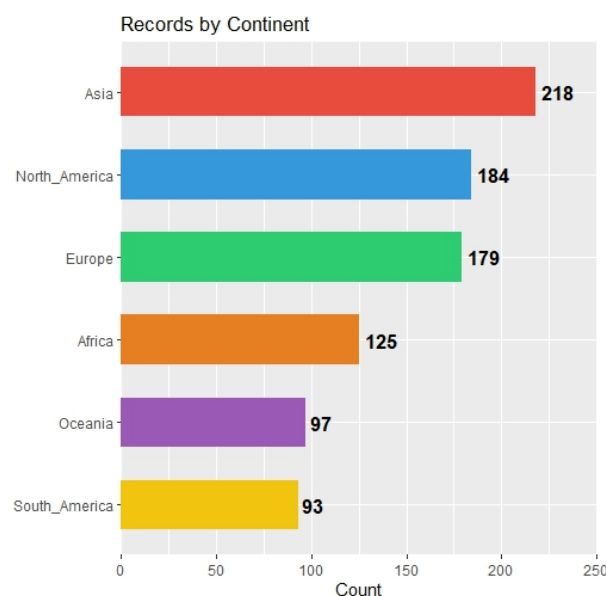


Figure 4.18: Distribution of Virus Records Across Continents

Figure 4.18 presents the distribution of virus-related records across different continents represented in the dataset.

The dataset shows broad geographical coverage across six continents, reflecting the global distribution of arboviruses and arthropod-specific viruses. The continental distribution provides important insight into the ecological and epidemiological patterns associated with vector-borne viral diseases under diverse environmental and climatic conditions.

Among all continents, Asia contributes the highest number of observations with 218 records. The large representation of Asia may be due to its high population density, tropical and subtropical climatic conditions, extensive biodiversity, and favorable environments for mosquito and tick survival. Many emerging and re-emerging arboviral diseases such as Dengue fever, Japanese encephalitis, and Zika virus infections are highly prevalent in several Asian countries.

North America represents the second most dominant continent with 184 observations, followed closely by Europe with 179 observations. These regions maintain strong virus surveillance systems and extensive epidemiological monitoring programs, adding to the higher availability of recorded virus data. In addition, climate change and expanding vector habitats have increased the importance of vector-borne diseases within temperate regions.

Africa contributes 125 observations and remains epidemiologically significant due to the endemic presence of several mosquito-borne and tick-borne viral infections. Warm temperatures, seasonal rainfall, and ecological diversity create highly favorable conditions for vector spread and zoonotic transmission cycles.

Oceania and South America contribute 97 and 93 observations respectively. These regions are also recognized for substantial arboviral activity, particularly involving mosquito-borne viruses transmitted by *Aedes* and *Culex* species. Tropical rainforest environments and humid climatic conditions further support vector survival and virus circulation.

The continental variation observed in the dataset may depend on several interconnected factors including climate, vector abundance, biodiversity, public health infrastructure, urbanization, land-use patterns, and disease surveillance efficiency. Tropical and subtropical regions generally show higher vulnerability to vector-borne diseases because environmental conditions strongly favor arthropod breeding and virus replication.

Overall, the distribution shows that vector-borne viral diseases are a major global public health concern affecting multiple continents. The widespread geographical representation emphasizes the importance of international surveillance systems, climate-based risk assessment, vector control programs, and collaborative epidemiological research for understanding and controlling emerging infectious diseases.

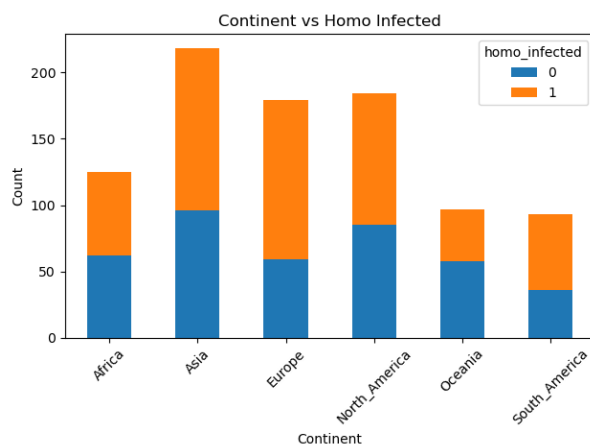


Figure 4.19: Relationship Between Continent and Human Infectivity

4.6.14 Host Groups

The dataset contains multiple vertebrate host groups that play important roles in the transmission, maintenance, amplification, and ecological circulation of Arboviruses and Arthropod-Specific Viruses. These host groups include birds, mammals, reptiles, and amphibians, each with distinct epidemiological roles in virus persistence and zoonotic spillover. The statistical distribution of host-related observations provides important insight into host susceptibility, infection prevalence, ecological adaptation, and transmission dynamics.

Humans (*Homo sapiens*) represent the most dominant host category in the dataset with a very high mean observation value of 667.7 and a maximum frequency of 11510. This strong dominance shows that the dataset is heavily focused on viruses associated with human infection and public health importance. Human infections caused by arboviruses may produce febrile illness, encephalitis, hemorrhagic fever, neurological complications, congenital abnormalities, and epidemic outbreaks. Viruses such as Dengue virus, Zika virus, Chikungunya virus, Japanese encephalitis virus, and West Nile virus are strongly associated with human infection through mosquito vectors.

The class *Aves* (birds) also shows substantial epidemiological importance with a mean value of 1115.4 and a maximum observation frequency of 1671. Birds function as major reservoir and amplification hosts for several mosquito-borne viruses. Migratory bird species play a critical role in long-distance virus spread across geographical regions and continents. Important arboviruses such as West Nile virus and Japanese encephalitis virus commonly circulate within bird-mosquito transmission cycles.

The order *Artiodactyla*, which includes cattle, goats, pigs, sheep, and deer, represents another highly important host group with a mean value of 97.35 and a maximum observation count of 5030. These domesticated and wild ungulates frequently act as amplification hosts and reservoir hosts for several zoonotic viral pathogens. Viral infections in livestock populations are epidemiologically important because they increase opportunities for vector-mediated transmission and zoonotic spillover into humans.

Rodents belonging to the order *Rodentia* show a mean observation value of 16.71 and a maximum frequency of 146. Rodents are ecologically significant reservoir hosts because of their rapid reproduction, environmental adaptability, and close association with human settlements. Their ability to maintain zoonotic pathogens within natural ecosystems plays a major role in virus persistence and transmission.

The order *Perissodactyla*, which includes horses and related species, shows a mean observation value of 10.9 with a maximum frequency of 164. Horses are particularly important sentinel hosts for several arboviruses including Venezuelan equine encephalitis virus and West Nile virus. Monitoring equine infections often provides early epidemiological warning of virus outbreaks.

Primates (*Primates*) show a mean observation value of 3.627 and a maximum frequency of 195. Non-human primates are biologically important because of their close genetic similarity to humans. They frequently take part in sylvatic transmission cycles involving mosquito vectors and serve as amplification hosts for zoonotic pathogens.

The order *Carnivora* has a mean value of 2.706 and a maximum observation frequency of 211. Carnivorous mammals such as dogs, foxes, raccoons, and wild carnivores may function as reservoir hosts, secondary hosts, or ecological carriers of several viral pathogens.

Bat species belonging to the order *Chiroptera* show a mean observation frequency of

2.039 with a maximum value of 19. Bats are globally recognized as important natural reservoirs for emerging viral pathogens because of their unique immune adaptations and ability to carry viruses without severe clinical disease. Their migration behavior and colony structure facilitate virus circulation and interspecies transmission.

Several less represented host groups also add to the ecological diversity of the dataset. *Eulipotyphla* shows a mean value of 0.3962, while *Reptilia* contributes 0.2578. Amphibians belonging to the order *Anura* show a mean value of 0.615, whereas *Didelphimorphia*, *Lagomorpha*, *Pilosa*, and *Diprotodontia* show lower frequencies. Although these host groups add fewer observations, they expand the ecological complexity and biological diversity of host-vector-virus interactions represented in the dataset.

The dominance of humans, birds, domesticated mammals, and rodents shows that zoonotic transmission and host-vector interactions are central components of arboviral epidemiology. Different host groups have distinct ecological behaviors, immune responses, habitat preferences, migration patterns, and susceptibility levels that directly influence virus replication, mutation, adaptation, transmission efficiency, and outbreak potential.

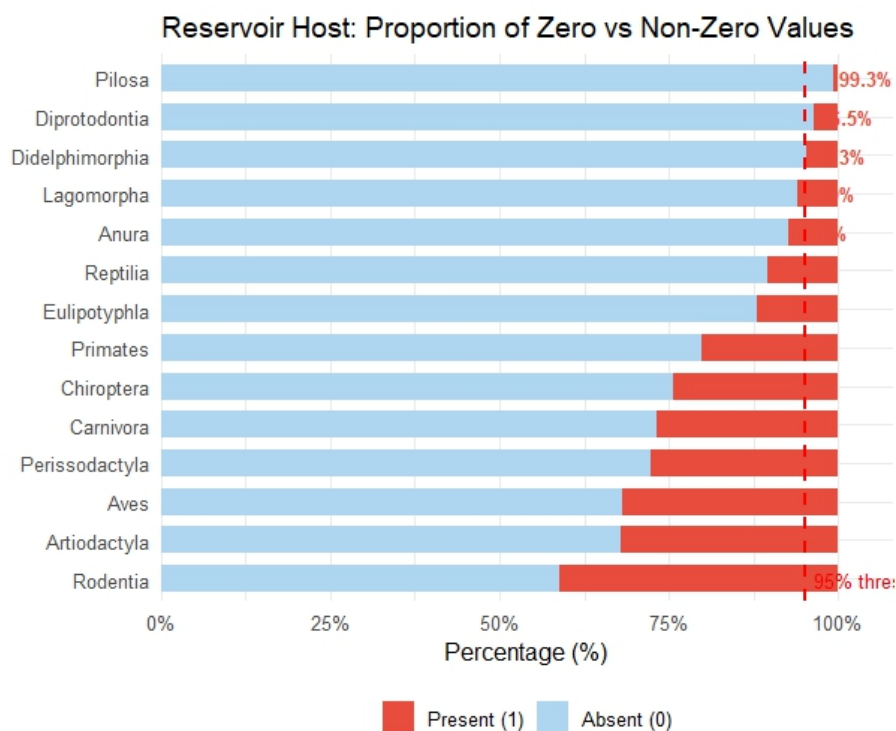


Figure 4.20: Proportion of Zero and Non-Zero Values in Reservoir Host Variables

Figure 4.20 shows the proportion of zero (absence) and non-zero (presence) values across different reservoir host variables included in the dataset. The analysis shows that most host-related variables are highly sparse, meaning that the majority of observations contain zero values.

The dominance of zero values shows that most viruses are associated with only a limited number of specific host groups rather than being broadly distributed across all host taxa. This pattern is biologically expected because many arboviruses and arthropod-associated viruses show strong host specificity and ecological adaptation.

Among all host groups, *Rodentia* shows the highest proportion of non-zero observations, showing that rodents play a major role as reservoir hosts within the dataset.

Rodents are epidemiologically important because of their rapid reproduction, ecological adaptability, and close association with human environments, which facilitate virus persistence and zoonotic transmission.

The host groups *Artiodactyla* and *Aves* also show higher proportions of non-zero values. Birds frequently function as natural reservoir hosts and amplification hosts for mosquito-borne viruses such as West Nile virus and Japanese encephalitis virus, while domesticated ungulates belonging to *Artiodactyla* play a major role in zoonotic transmission cycles and livestock-associated viral infections.

Moderate levels of non-zero observations are seen for *Perissodactyla*, *Carnivora*, *Chiroptera*, and *Primates*. These host groups include horses, carnivorous mammals, bats, and non-human primates, all of which are epidemiologically significant because they can maintain virus circulation or contribute to spillover transmission under suitable ecological conditions.

Several host variables including *Reptilia*, *Eulipotyphla*, *Anura*, *Lagomorpha*, *Pilosa*, *Diprotodontia*, and *Didelphimorphia* contain very high proportions of zero values with only a few non-zero observations. This suggests that these hosts take part only occasionally in virus maintenance or transmission pathways and are associated with relatively specialized ecological niches.

From a statistical and machine learning perspective, the strong dominance of zero values shows the presence of sparse high-dimensional data. Sparse variables may add limited predictive information and can introduce noise, instability, or overfitting during model development. Therefore, variable selection methods such as LASSO regularization, Ridge regression, or dimensionality reduction techniques may be useful for identifying the most epidemiologically important host variables.

Biologically, the sparsity seen in host variables reflects the ecological specialization of viruses and the selective interaction between pathogens, hosts, and arthropod vectors. Virus transmission depends strongly on host susceptibility, vector feeding preference, environmental conditions, and ecological compatibility.

Overall, the zero versus non-zero distribution highlights that only a limited set of host groups plays a major role in virus ecology and zoonotic transmission. The dominance of rodents, birds, humans, and domesticated mammals emphasizes their major epidemiological importance in the maintenance and spread of vector-borne viral diseases.

4.6.15 Distribution of Virus Counts

The variable **Count**, scientifically interpreted as **Virus Occurrence Frequency** or **Environmental Detection Intensity**, represents the total number of times a particular virus has been detected, isolated, or reported within different ecological, geographical, and climatic environments. This variable provides important quantitative information about the environmental prevalence, transmission intensity, and epidemiological importance of viral pathogens.

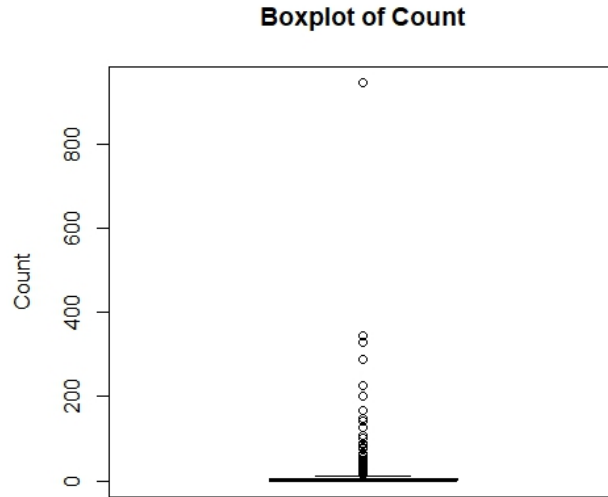


Figure 4.21: Boxplot of Virus Detection Counts

Figure 4.21 presents the boxplot distribution of virus detection counts. The figure clearly shows the presence of strong positive skewness and many extreme outliers within the dataset. Most virus detection counts remain concentrated near the lower range, whereas only a few viruses show exceptionally high occurrence frequencies.

The descriptive statistics further support this observation. The minimum value of the variable is 1, the first quartile (Q_1) is 1, the median is 3, and the third quartile (Q_3) is 6. However, the maximum value reaches 944, showing the existence of extremely high-frequency virus observations. The mean value (9.337) is considerably larger than the median, confirming the presence of a highly right-skewed distribution.

The boxplot shows many outliers above the upper whisker, suggesting that only a few viruses dominate the ecological transmission landscape. This type of statistical behavior closely resembles a **Power-law Distribution** or heavy-tailed distribution, which is commonly seen in ecological and epidemiological systems where a small number of highly successful pathogens account for a disproportionately large number of transmission events.

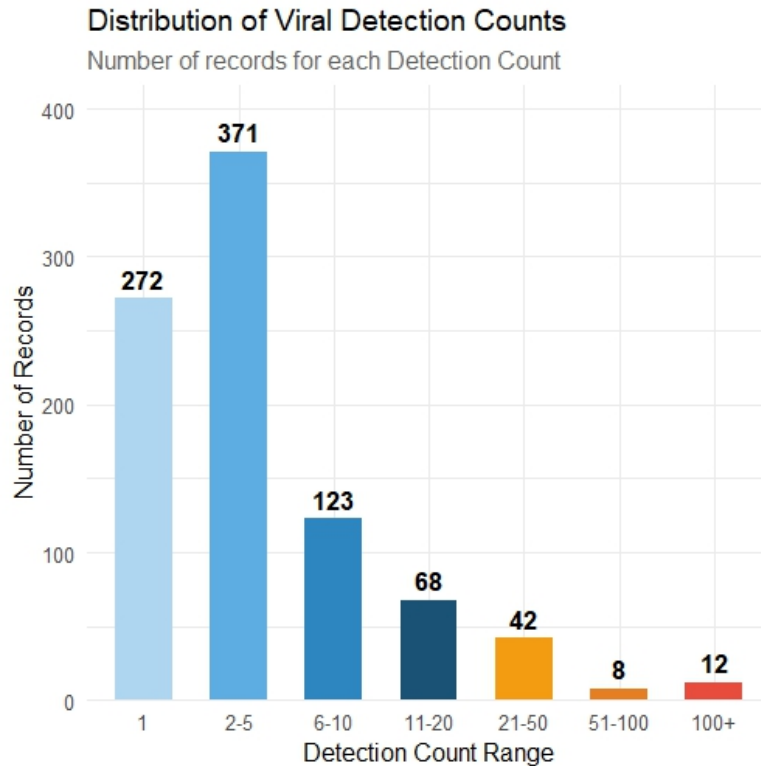


Figure 4.22: Distribution of Virus Detection Count Categories

Figure 4.22 shows the grouped frequency distribution of virus detection counts across different count intervals. The distribution shows that most viruses occur at relatively low frequencies, while only a small set shows very high detection frequencies.

Among all categories, the interval 2–5 detections contains the highest number of records with 371 observations, followed by single detections (1) with 272 observations. The interval 6–10 contributes 123 observations, while 11–20 contributes 68 observations. Higher detection ranges such as 21–50, 51–100, and 100+ contain progressively fewer observations, with only 42, 8, and 12 records respectively.

The dominance of low-frequency virus detections shows that most viruses remain relatively rare within ecological transmission systems. Many viruses may circulate only within restricted host populations, localized ecological niches, or specific geographical regions. In contrast, only a few highly adaptive viruses achieve extensive environmental prevalence and sustained transmission.

From an epidemiological perspective, viruses with very high detection frequencies are generally associated with stronger ecological adaptability, broader host range, efficient vector transmission, and increased zoonotic potential. High-frequency viruses often circulate continuously within vector-host ecosystems and therefore pose substantial public health concern because of their elevated spillover risk into human populations.

Viruses such as West Nile virus, Japanese encephalitis virus, Dengue virus, Zika virus, and Tick-borne encephalitis virus are likely contributors to the upper detection categories because of their widespread geographical distribution and major epidemiological importance.

The strong imbalance between low-frequency and high-frequency viruses also has important statistical implications. The heavy-tailed distribution and extreme outliers may violate normality assumptions required in classical statistical models. Consequently, log-

arithmetic transformation and regularization approaches such as LASSO and Ridge regression become essential for stabilizing variance, reducing outlier influence, improving model interpretability, and enhancing predictive performance.

Furthermore, virus detection frequency acts as an indirect measure of environmental viral burden and infection pressure. Greater environmental prevalence increases opportunities for host-vector interaction, virus replication, mutation, ecological adaptation, and zoonotic transmission into human populations.

Overall, both the boxplot and grouped frequency distribution show that the ecological transmission landscape is highly unequal, where only a few dominant viruses contribute disproportionately to the global burden of vector-borne viral diseases. These findings emphasize the importance of continuous surveillance, ecological monitoring, and predictive epidemiological modeling for identifying emerging high-risk pathogens and controlling future outbreaks.

4.6.16 Vector Diversity

To systematically evaluate the ecological adaptability, transmission flexibility, and zoonotic potential of the studied viruses, a set of cumulative vector diversity indices was constructed. These indices measure the taxonomic breadth of arthropod vectors associated with each virus across multiple hierarchical biological levels, ranging from individual vector species to broader taxonomic classes.

The main goal of constructing these variables is to understand how broadly a virus can use different vector groups within natural ecosystems. Viruses capable of infecting or circulating through taxonomically diverse vectors are generally considered ecologically flexible and epidemiologically important because they have greater environmental resilience, broader geographical distribution, and enhanced zoonotic spillover potential.

The taxonomic hierarchy used in this study follows the standard biological classification system:

$$\text{Species} \rightarrow \text{Genus} \rightarrow \text{Family} \rightarrow \text{Order} \rightarrow \text{Class}$$

Each higher taxonomic level reflects increasing ecological generalization and evolutionary diversity.

1. Species-Level Diversity (`total_vector`)

The variable `total_vector` represents the total number of unique arthropod vector species associated with each virus.

$$total_vector_i = |S_i| \tag{4.1}$$

where S_i is the set of distinct vector species linked to virus i .

2. Genus-Level Diversity (`vector_G_T`)

The variable `vector_G_T` measures the number of unique vector genera associated with a virus.

$$vector_G_T_i = |G_i| \tag{4.2}$$

where G_i represents the set of distinct vector genera.

3. Family-Level Diversity (vector_F_T)

The variable `vector_F_T` measures the number of unique vector families associated with each virus.

$$vector_F_T_i = |F_i| \quad (4.3)$$

where F_i is the set of distinct vector families.

4. Order-Level Diversity (vector_O_T)

The variable `vector_O_T` measures vector diversity at the order level.

$$vector_O_T_i = |O_i| \quad (4.4)$$

where O_i is the set of distinct vector orders.

5. Class-Level Diversity (vector_C_T)

The variable `vector_C_T` represents the highest level of vector taxonomic diversity included in the study.

$$vector_C_T_i = |C_i| \quad (4.5)$$

where C_i is the set of distinct vector classes.

To show how these diversity indices are calculated, consider a hypothetical example involving West Nile Virus (WNV). Suppose the virus is detected in the following vector species:

- *Culex pipiens*
- *Culex tarsalis*
- *Aedes aegypti*
- *Ixodes ricinus*

The resulting diversity variables are calculated as follows:

$$total_vector = 4 \quad (4.6)$$

$$vector_G_T = 3 \quad (Culex, Aedes, Ixodes) \quad (4.7)$$

$$vector_F_T = 3 \quad (4.8)$$

$$vector_O_T = 2 \quad (Diptera, Ixodida) \quad (4.9)$$

$$vector_C_T = 2 \quad (Insecta, Arachnida) \quad (4.10)$$

This example shows how a single virus may bridge fundamentally different ecological transmission systems involving both mosquito-borne and tick-borne pathways.

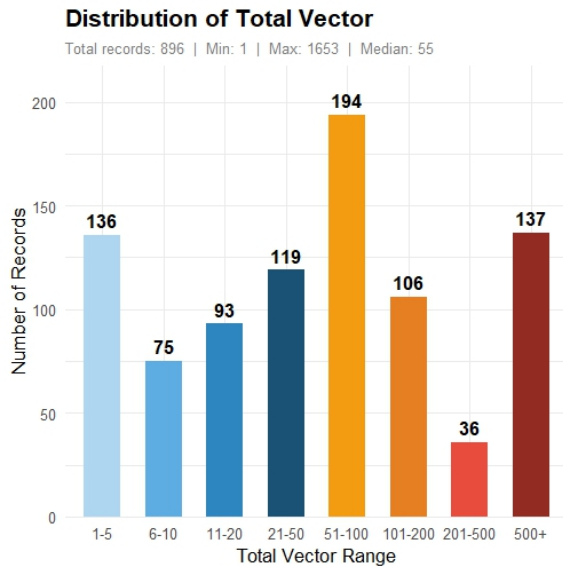


Figure 4.23: Distribution of Species-Level Vector Diversity (**total_vector**)

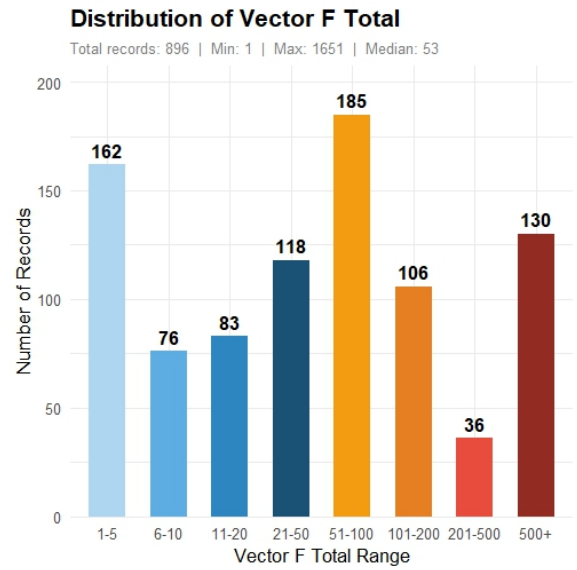


Figure 4.24: Distribution of Family-Level Vector Diversity (**vector_F_T**)

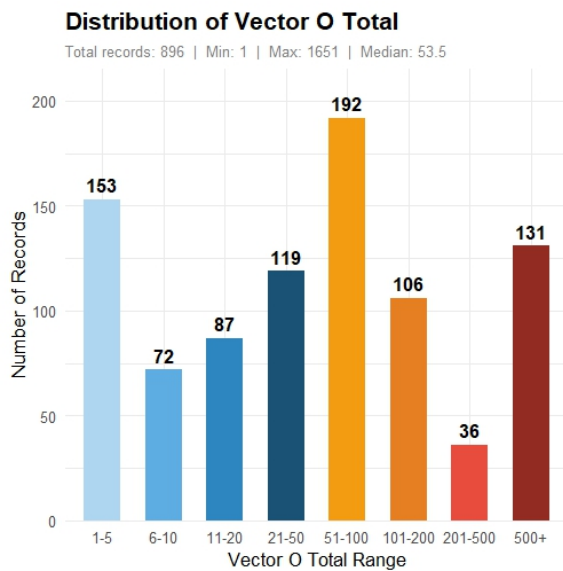


Figure 4.25: Distribution of Order-Level Vector Diversity (**vector_O_T**)

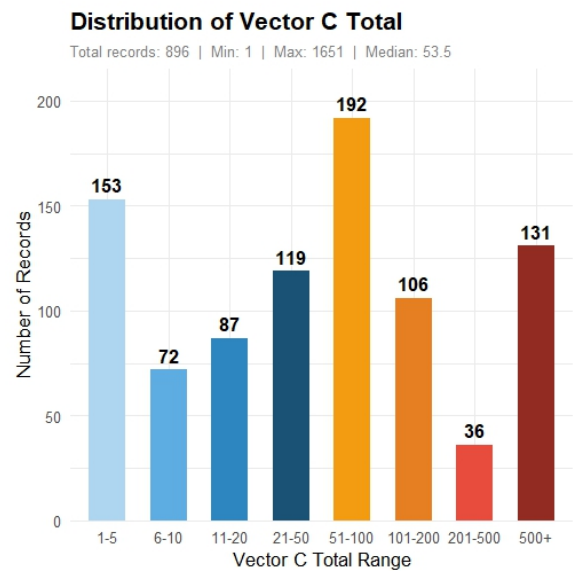


Figure 4.26: Distribution of Class-Level Vector Diversity (**vector_C_T**)

4.7 Cross-Vector Variables

Beyond the simple count of vector species, the ability of a virus to cross large phylogenetic boundaries among its associated vectors is an important measure of ecological flexibility and evolutionary adaptability. To systematically measure this phenomenon, a set of Cross-Vector variables was developed.

These variables measure the diversity of virus transmission pathways across multiple hierarchical taxonomic levels, including Genus, Family, Order, and Class. Together, they provide important ecological and epidemiological insight into the adaptability, transmission complexity, and zoonotic emergence potential of viruses.

4.7.1 Calculation of Cross-Vector Variables

For each virus i , the cross-taxa breadth variables were calculated as the total number of unique taxonomic units identified within its associated vector range.

The variable `cross_vector_G` measures the number of unique arthropod vector genera capable of transmitting virus i .

$$cross_vector_G_i = |G_i| \quad (4.11)$$

where G_i is the set of distinct vector genera.

The variable `cross_vector_F` represents the number of distinct vector families associated with virus i .

$$cross_vector_F_i = |F_i| \quad (4.12)$$

where F_i is the set of distinct vector families.

The variable `cross_vector_O` measures diversity at the order level.

$$cross_vector_O_i = |O_i| \quad (4.13)$$

where O_i is the set of distinct vector orders.

The variable `cross_vector_C` represents the broadest measure of vector plasticity.

$$cross_vector_C_i = |C_i| \quad (4.14)$$

where C_i is the set of distinct vector classes.

Particularly, a value of:

$$cross_vector_C > 1$$

shows that a virus is associated with more than one vector class, suggesting the potential to bridge different ecological transmission systems. For example, consider a virus associated with the following vectors:

- *Culex pipiens*
- *Aedes aegypti*
- *Ixodes ricinus*

The corresponding cross-taxa breadth variables would be:

$$cross_vector_G = 3 \quad (4.15)$$

$$cross_vector_F = 3 \quad (4.16)$$

$$cross_vector_O = 2 \quad (4.17)$$

$$cross_vector_C = 2 \quad (4.18)$$

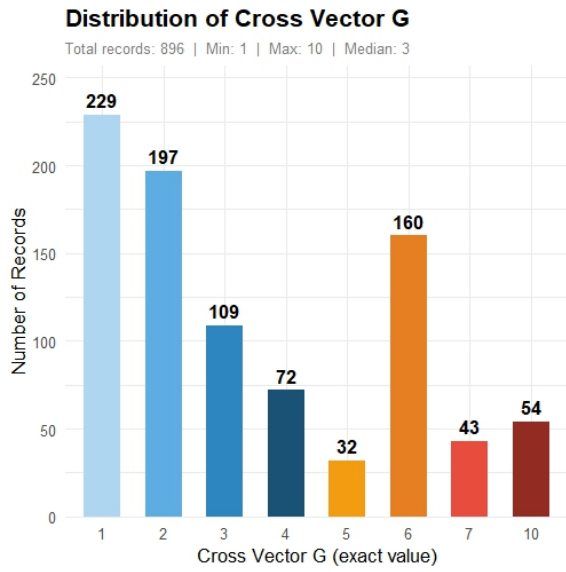


Figure 4.27: Distribution of Genus-Level Cross-Taxa Breadth (`cross_vector_G`)

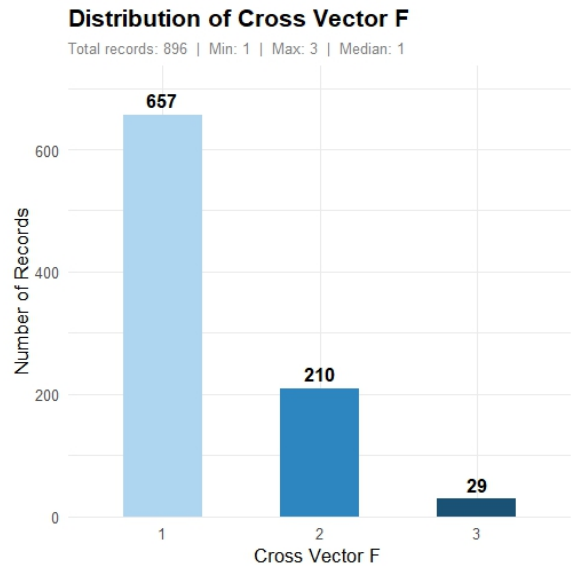


Figure 4.28: Distribution of Family-Level Cross-Taxa Breadth (`cross_vector_F`)

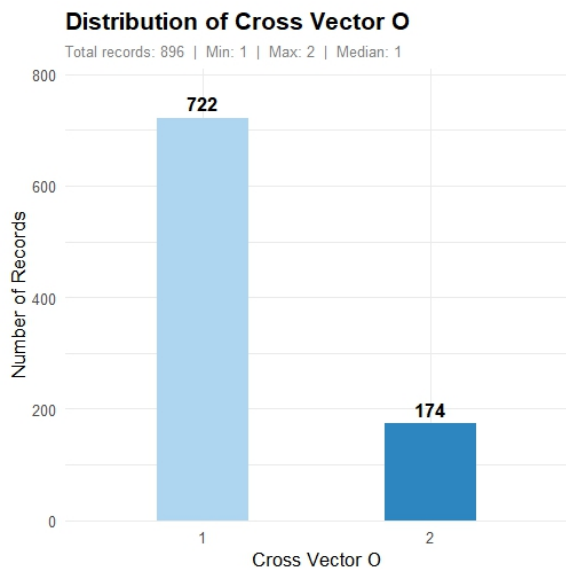


Figure 4.29: Distribution of Order-Level Cross-Taxa Breadth (`cross_vector_O`)

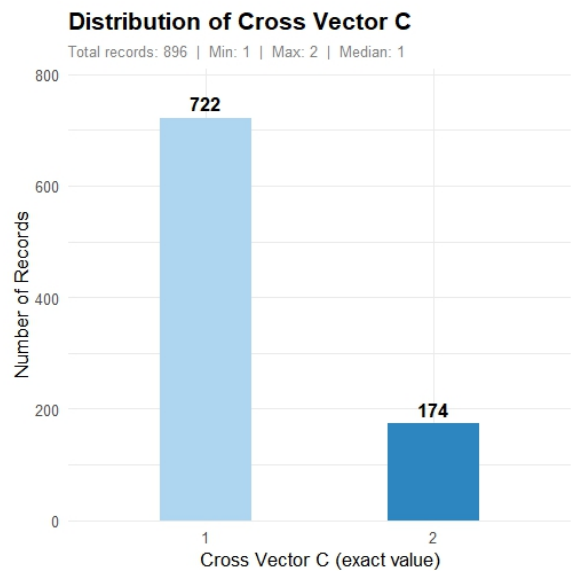


Figure 4.30: Distribution of Class-Level Cross-Taxa Breadth (`cross_vector_C`)

4.8 cross_host

The variable `cross_host` was developed to measure the ecological host breadth and vertebrate adaptability of viral pathogens. This variable represents the number of distinct vertebrate host orders associated with a particular virus and serves as an important ecological measure of virus plasticity.

Unlike individual host variables that describe specific reservoir groups, the `cross_host` variable provides a broader ecological summary of host diversity across the entire transmission system. Therefore, this variable captures the overall complexity of host-virus interaction networks rather than isolated host associations.

The `cross_host` variable was calculated using vertebrate animal orders recorded in the dataset, including groups such as:

- *Aves* (birds)
- *Rodentia* (rodents)
- *Chiroptera* (bats)
- *Carnivora* (carnivorous mammals)
- *Primates*
- *Artiodactyla*
- and several other vertebrate groups.

For each virus, the number of unique host orders with virus presence was counted to generate the final `cross_host` value. Thus, higher values show broader host adaptability and greater ecological flexibility.

Mathematically, the variable was derived using a binary presence–absence approach:

$$cross_host_i = \sum_{j=1}^{14} I(C_{i,j} > 0) \quad (4.19)$$

where:

$$I(C_{i,j} > 0) = \begin{cases} 1 & \text{if } C_{i,j} > 0 \\ 0 & \text{if } C_{i,j} = 0 \end{cases} \quad (4.20)$$

This formulation ensures that the variable measures ecological diversity rather than raw sampling intensity.

Biologically, viruses with low `cross_host` values are generally restricted to narrow ecological niches and specific host systems. Such pathogens are considered ecological specialists and often have lower spillover potential.

In contrast, viruses with high `cross_host` values can infect multiple taxonomically distinct vertebrate groups. This broader host adaptability greatly increases opportunities for:

- cross-species transmission,
- environmental persistence,

- ecological expansion,
- and zoonotic spillover into human populations.

From an epidemiological perspective, broader host diversity increases the probability of human exposure because viruses circulating across multiple wildlife reservoirs interact with more diverse ecological environments and transmission pathways.

Figure 4.31 shows the distribution of the `cross_host` variable across virus observations in the dataset. The figure shows that most viruses have relatively narrow host breadth, whereas only a limited set has very broad vertebrate host adaptability. Such highly adaptable viruses are more likely to show stronger ecological persistence, wider geographical distribution, and increased human infection potential.

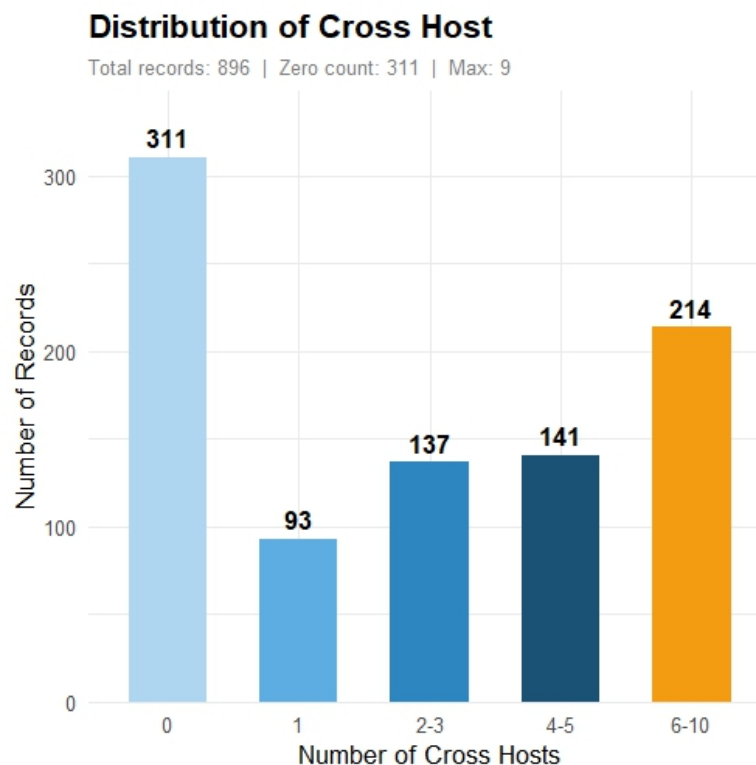


Figure 4.31: Distribution of Host Breadth Variable (`cross_host`)

The relationship between cross-host diversity (`cross_host`) and human infectivity status (`homo_infected`) is shown in Figure 4.32. The boxplot compares the distribution of cross-host diversity between human-infecting and non-human-infecting viruses.

The figure shows a large difference in host diversity between the two groups. Non-human-infecting viruses are mainly concentrated near zero, showing very limited host range diversity. In contrast, human-infecting viruses show considerably higher cross-host diversity, with a wider distribution and a median value much greater than that of the non-human-infecting group.

This pattern suggests that viruses capable of infecting humans tend to infect a broader range of host species, thereby increasing the likelihood of cross-species transmission and zoonotic spillover. The larger variability seen among human-infecting viruses also shows greater ecological adaptability and transmission potential.

The visual observation strongly supports the results from the Mann–Whitney U test, which showed a highly significant difference between the two groups ($U = 3082.0$, $p < 0.001$). This finding shows that host diversity plays a major role in virus pathogenicity and zoonotic transmission dynamics.

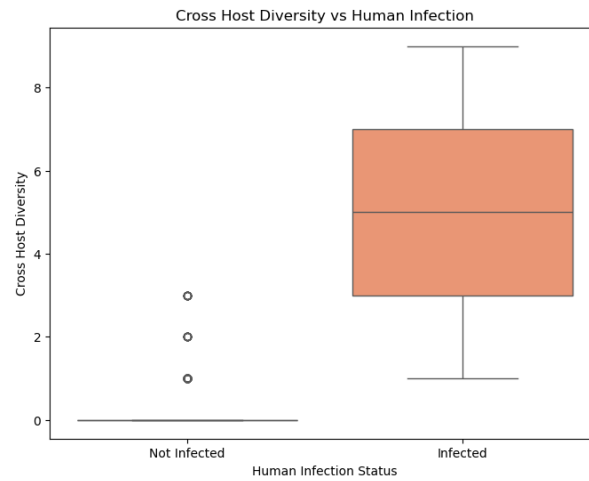


Figure 4.32: Relationship Between Cross-Host Diversity and Human Infectivity

4.8.1 Chi-Square Test Between Virus Group and Human Infectivity

We carried out a Chi-Square test of independence to examine the association between `Virus_Group` and human infectivity status (`homo_infected`). The Chi-Square test was chosen because both variables are categorical. The test checks whether the observed frequencies differ significantly from the expected frequencies under the assumption of independence.

The contingency table from the analysis is shown below:

Table 4.1: Contingency Table of Virus Group and Human Infectivity

Virus Group	Non-Human Infecting (0)	Human Infecting (1)
Arbovirus	268	500
Arthropod-specific virus	128	0

The Chi-Square test gave the following results:

$$\chi^2 = 185.92$$

$$p < 0.001$$

with 1 degree of freedom.

Since the p-value was much smaller than the significance level of 0.05, we rejected the null hypothesis of independence. Therefore, a statistically significant association exists between virus ecological group and human infectivity status.

The results show that Arboviruses are much more associated with human infection compared to arthropod-specific viruses. This finding suggests that virus ecological classification plays an important role in zoonotic spillover and transmission dynamics.

4.8.2 Chi-Square Test Between Continent and Human Infectivity

We carried out a Chi-Square test of independence to evaluate the association between geographical continent distribution and human infectivity status (`homo_infected`). The Chi-Square test was chosen because both variables are categorical. The analysis compared the observed frequencies with the expected frequencies under the assumption of independence.

The contingency table from the analysis is shown below:

Table 4.2: Contingency Table of Continent and Homo infection

Continent	Non-Human Infecting (0)	Human Infecting (1)
Africa	62	63
Asia	96	122
Europe	59	120
North America	85	99
Oceania	58	39
South America	36	57

The Chi-Square test gave the following results:

$$\chi^2 = 21.65$$

$$p < 0.001$$

with 5 degrees of freedom.

Since the p-value was smaller than the significance level of 0.05, we rejected the null hypothesis of independence. Therefore, a statistically significant association exists between continent and human infectivity status.

The results suggest that the geographical distribution of viruses varies significantly with respect to human infection capability. In particular, Europe, Asia, and South America showed relatively higher frequencies of human-infecting viruses, showing that ecological and geographical factors may play a role in zoonotic spillover dynamics and virus transmission patterns.

4.8.3 Mann–Whitney U Test Between Cross-Host Diversity and Human Infectivity

We carried out a Mann–Whitney U test to compare the distribution of `cross_host` between human-infecting and non-human-infecting viruses. The Mann–Whitney U test was chosen because `cross_host` is a non-normally distributed numerical variable, while `homo_infected` is a binary categorical variable.

The hypotheses for the test were:

H_0 : The distribution of `cross_host` is identical across `homo_infected` groups

H_1 : The distribution of `cross_host` differs between `homo_infected` groups

The test gave the following results:

$$U = 3082.0$$

$$p < 0.001$$

Since the p-value was much smaller than the significance level of 0.05, we rejected the null hypothesis. Therefore, a statistically significant difference exists in cross-host diversity between human-infecting and non-human-infecting viruses.

4.8.4 Mann–Whitney U Test Between Total Vector Diversity and Human Infectivity

We carried out a Mann–Whitney U test to compare the distribution of `total_vector` between human-infecting and non-human-infecting viruses. The Mann–Whitney U test was chosen because `total_vector` is a non-normally distributed numerical variable, while `homo_infected` is a binary categorical variable.

The hypotheses for the analysis were:

H_0 : The distribution of total_vector is identical across homo_infected groups

H_1 : The distribution of total_vector differs between homo_infected groups

The test gave the following results:

$$U = 34646.5$$

$$p < 0.001$$

Since the p-value was much smaller than the significance level of 0.05, we rejected the null hypothesis. Therefore, a statistically significant difference exists in total vector diversity between human-infecting and non-human-infecting viruses.

The findings suggest that viruses capable of infecting humans are associated with significantly different vector abundance and diversity patterns. This shows that vector complexity may play a major role in zoonotic spillover dynamics and virus transmission potential.

Chapter 5

Data Processing

5.1 Variable Selection for Analysis

The original dataset contained 39 variables, providing a highly granular view of virus ecology. However, for improved machine learning performance, a systematic variable selection process was conducted. The primary objective was to achieve model parsimony while preserving important biological information and reducing mathematical redundancies such as multicollinearity.

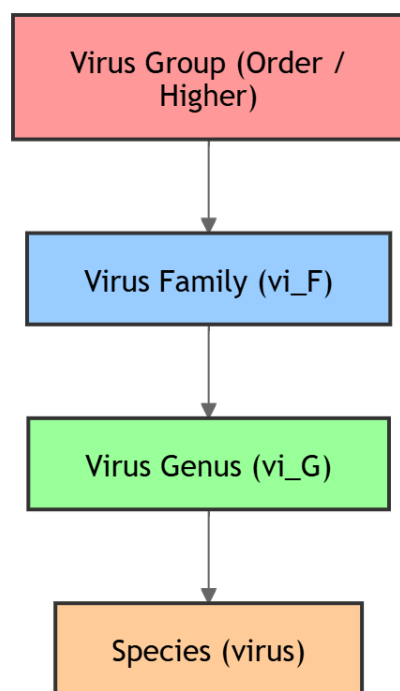


Figure 5.1: Hierarchical structure of virus-related variables, illustrating the taxonomic levels from Virus Group down to individual family classification.

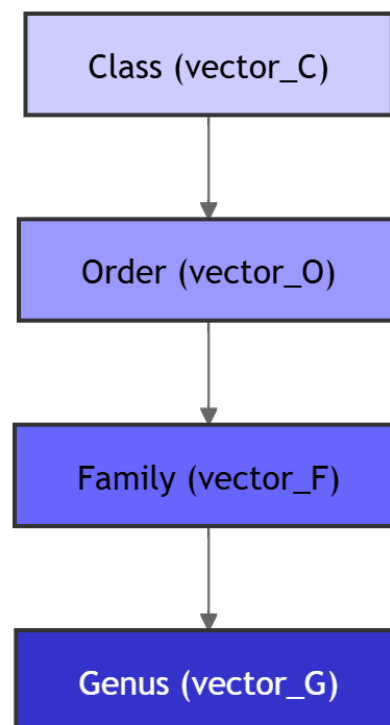


Figure 5.2: Hierarchical structure of vector-related variables, showing taxonomic levels from vector class down to genus and aggregated diversity counts.

5.1.1 Prevention of Data Leakage

The variables `homo` (total human cases) and `Count` (raw detection frequency) were flagged for potential exclusion. Because `homo` is a direct numerical representation of the target variable (`homo_infected`), its inclusion would lead to *data leakage*, resulting in a model that is mathematically trivial and scientifically invalid. A final decision on these variables is deferred to Section 5.1.6, where their exclusion is formally justified.

5.1.2 Mitigation of Multicollinearity

Redundant taxonomic levels (e.g., `vi_G`, `vector_G`, `vector_D`, `vector_C`) and nested vector total counts (`vector_G_T`, `vector_F_T`, etc.) were removed. Statistical testing showed high correlation between these variables. Retaining them would violate the assumption of independence required for Generalized Linear Models (GLM) and destabilize coefficient estimation in LASSO and Ridge regressions.

5.1.3 Addressing Data Sparsity

Minor host groups such as `Reptilia` and `Anura` were excluded due to extreme data sparsity, with over 95% of observations containing zero values. Such variables introduce significant noise without contributing meaningful predictive power, potentially leading to model overfitting.

Variables such as `Pilosa` and `Lagomorpha` also exhibited high levels of sparsity; however, they were evaluated for exclusion but retained in the final predictor set due to their marginal biological relevance as occasional vertebrate reservoirs in zoonotic transmission cycles. Their retention was further justified by the fact that regularization techniques (LASSO and Ridge) would automatically shrink their coefficients toward zero if they provided limited predictive information — a behavior confirmed in the final coefficient estimates, where both variables showed near-zero coefficients.

5.1.4 Variable Construction and Dimensionality Reduction

To further optimize the data for machine learning, two major variable construction steps were implemented:

- **Taxonomic Grouping:** Virus families with occurrence frequency below 5% were grouped into a “Rare Family” category to ensure statistical robustness in each group.
- **Climatic Grouping:** The Köppen climate classification codes were reduced to their primary categories (A, B, C, D, E). This preserved the essential environmental signal while reducing the degrees of freedom required by the model.

5.1.5 Candidate Variable Set After Multicollinearity Assessment

A multicollinearity assessment was performed using the Pearson correlation matrix to improve model robustness, interpretability, and predictive stability. Variables with pairwise correlation:

$$|r| \geq 0.75$$

were considered highly collinear and selectively removed based on biological redundancy and statistical dependency.

Figure 5.3 presents the full Pearson correlation heatmap with variable pairs exceeding the $|r| \geq 0.75$ threshold explicitly highlighted. Following removal of these highly collinear variables, the resulting reduced correlation heatmap is shown in Figure 5.4.

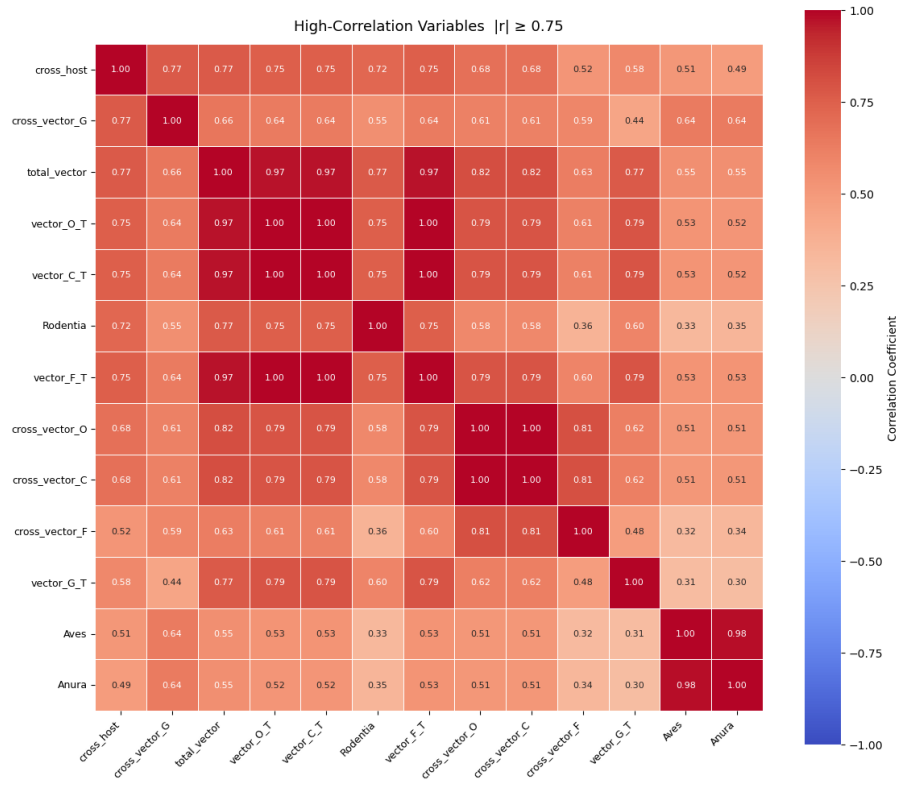


Figure 5.3: Heatmap with variable pairs exceeding the $|r| \geq 0.75$ collinearity threshold highlighted.

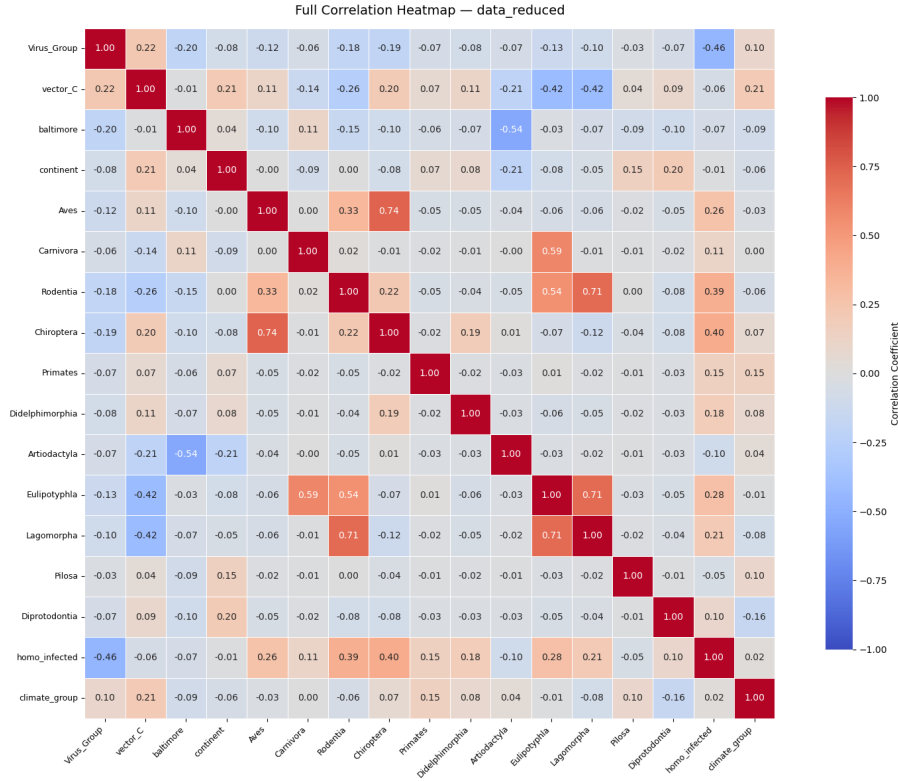


Figure 5.4: Correlation heatmap after removal of variables exceeding the $|r| \geq 0.75$ threshold.

After this initial filtering, the following **candidate** variables were retained for further evaluation. This represents an *intermediate* selection stage; additional exclusions based on data leakage and linear redundancy are applied in Section 5.1.6.

- Virus_Group
- vector_C
- Count[†]
- baltimore
- continent
- Aves
- Carnivora
- Rodentia
- Chiroptera
- Primates
- homo[†]
- Didelphimorphia
- Artiodactyla
- Eulipotyphla
- Lagomorpha
- Pilosa
- Diprotodontia
- cross_host[†]
- cross_vector_F[†]
- climate_group

[†] Variables marked for further exclusion in Section 5.1.6.

5.1.6 Refined Variable Exclusion Criteria

To ensure the mathematical validity of the predictive models and enhance biological interpretability, a specific set of variables from the candidate set was excluded from the final predictor set. The rationale for dropping `homo`, `Count`, `cross_host`, and `cross_vector_F` is detailed below:

- **Prevention of Data Leakage (`homo` and `Count`):** The primary target variable is `homo_infected`. The variable `homo` (raw count of human cases) is a direct numerical precursor to the target; its inclusion would provide the model with “future information,” leading to trivial accuracy. Similarly, `Count` (viral detection frequency) often reflects surveillance intensity focused on known human pathogens, introducing “surveillance bias.”
- **Elimination of Linear Redundancy (`cross_host`):** While `cross_host` is an informative diversity index, it is mathematically derived from the presence of specific animal reservoirs (e.g., *Aves*, *Rodentia*, *Chiroptera*). Since this study retains individual high-signal animal orders as independent predictors, retaining `cross_host` would introduce direct linear dependency.
- **Mitigation of Multicollinearity (`cross_vector_F`):** Preliminary correlation analysis showed near-perfect multicollinearity between genus-level diversity (`cross_vector_G`) and family-level diversity (`cross_vector_F`). The family-level metric was dropped in favor of the genus-level metric, as the latter provides finer resolution of viral transmission flexibility.

5.2 Final Variable Set and Characterization

Following the two-stage selection process — initial multicollinearity filtering followed by refined exclusion of leakage and redundant variables — a **final set of 16 predictor variables** was retained. These variables represent the biological, taxonomic, and ecological dimensions of the global virome. Table 5.1 summarizes the final variable set and their respective data types.

Table 5.1: Final Predictor Variables and Data Types

Category	Variable Name	Data Type
Target Variable	<code>homo_infected</code>	Binary (1/0)
Virus Characteristics	<code>Virus_Group</code>	Categorical (Encoded)
	<code>baltimore</code>	Categorical (Encoded)
Vector Information	<code>vector_C</code>	Categorical (Encoded)
Ecological Reservoirs	<code>Aves</code> , <code>Carnivora</code> , <code>Rodentia</code>	Numerical (Counts)
	<code>Chiroptera</code> , <code>Primates</code>	Numerical (Counts)
	<code>Didelphimorphia</code> , <code>Artiodactyla</code>	Numerical (Counts)
	<code>Eulipotyphla</code> , <code>Lagomorpha</code>	Numerical (Counts)
	<code>Pilosa</code> , <code>Diprotodontia</code>	Numerical (Counts)
Geographic Context	<code>continent</code>	Categorical (Encoded)
Environmental Context	<code>climate_group</code>	Categorical (Encoded)

5.2.1 Scientific Rationale for the Final Set

The inclusion of these specific variables is based on their established roles in zoonotic spillover dynamics:

- **Host Reservoir Dynamics:** By retaining 11 distinct vertebrate orders (e.g., *Rodentia*, *Chiroptera*, *Aves*), the model captures specific risk profiles associated with different animal populations. Viruses with high detection counts in bats and rodents are significantly more likely to emerge as human pathogens.
- **Virus Taxonomy and Genetics:** The `baltimore` classification and `Virus_Group` allow the algorithms to identify genetic signatures (e.g., RNA mutation rates) that facilitate inter-species adaptation.
- **Ecological and Geographic Filters:** `climate_group` (major Köppen zones) and `continent` act as ecological filters determining the geographic range of vector-host interactions.
- **Vector Class Specificity:** Distinguishing between *Insecta* and *Arachnida* (via `vector_C`) provides crucial information on transmission mode and environmental stability.

5.3 Modeling Strategy

The analytical pipeline of this study is structured in two distinct stages, each serving a clearly defined purpose.

Stage 1 — Variable Selection and Coefficient Interpretation: LASSO Logistic Regression (L1 regularization) and Ridge Logistic Regression (L2 regularization) were applied to the 16-variable final set using a 70:30 stratified train-test split. These models were used primarily to identify the most biologically informative predictors through coefficient shrinkage and to assess the directional influence of ecological variables on human infectivity. They were not intended as the primary predictive classifiers.

Stage 2 — Predictive Classification: Random Forest and XGBoost classifiers were developed as the primary predictive models, trained on an 80:20 stratified split to maximize training data availability for ensemble learning. These two models are directly compared using accuracy, AUC, precision, recall, and F1-score. The different train-test ratios reflect the distinct purposes of regularized regression (interpretability) versus ensemble classification (predictive performance).

5.4 Motivation for Machine Learning Modeling

Predicting human infection potential requires understanding why certain viruses successfully infect humans while others remain restricted to animal or arthropod hosts. Although exploratory statistical analysis provides important biological insight, these approaches alone cannot capture the highly complex and non-linear ecological mechanisms underlying viral emergence.

5.4.1 Limitation of Simple Statistical Interpretation

Consider two hypothetical viruses:

- **Virus A** infects three closely related amphibian species.
- **Virus B** infects birds (*Aves*), rodents (*Rodentia*), and bats (*Chiroptera*).

In both cases $\text{cross_host} = 3$, yet their epidemiological significance is fundamentally different. Virus B must overcome dramatically different biological barriers — distinct immune systems, receptor structures, metabolic pathways, and transmission ecologies — demonstrating far greater ecological plasticity and zoonotic spillover potential.

This illustrates a critical biological principle:

Human infectivity is not determined solely by the number of infected hosts, but by the ecological diversity and biological complexity of those hosts.

Such relationships cannot be captured through simple linear statistics.

5.4.2 Why Machine Learning Was Necessary

The dataset presents several challenges that make classical statistical methods insufficient: high dimensionality, sparse zero-inflated host variables, hierarchical multicollinearity among vector taxonomy variables, and complex non-linear interactions among host, vector, climate, and geographic predictors.

Machine learning algorithms address these challenges by capturing non-linear transmission relationships, detecting hidden ecological interactions, and handling sparse high-dimensional data without strict distributional assumptions.

In this study, two complementary modeling strategies were employed:

- **LASSO and Ridge Regression** — for variable selection and coefficient interpretation, identifying the most biologically informative predictors through regularization.
- **Random Forest and XGBoost** — as primary predictive classifiers, capturing complex non-linear ecological interactions and producing final human infectivity predictions.

The ultimate goal is not merely statistical classification, but to uncover the hidden biological signatures of zoonotic emergence — enabling the transition from reactive outbreak surveillance toward proactive predictive epidemiology.

Chapter 6

LASSO Regression

6.1 Least Absolute Shrinkage and Selection Operator (LASSO)

6.1.1 Introduction

The Least Absolute Shrinkage and Selection Operator (LASSO), introduced by Tibshirani (1996), is a regularization technique designed to address two fundamental problems in regression analysis: overfitting and variable selection. Unlike Ordinary Least Squares (OLS), which minimizes only the residual sum of squares, LASSO imposes an L_1 penalty on the regression coefficients. This penalty simultaneously shrinks coefficient magnitudes toward zero and forces sufficiently weak coefficients to become exactly zero, thereby carrying out automatic variable selection.

Consider the usual regression setup with observations (y_i, x_{ij}) , where y_i is the i^{th} response and x_{ij} is the i^{th} observation of the j^{th} predictor, for $i = 1, 2, \dots, n$ and $j = 1, 2, \dots, p$. The linear regression model is

$$y_i = \beta_0 + \sum_{j=1}^p \beta_j x_{ij} + \varepsilon_i$$

The OLS estimators often have zero bias but large variance. By introducing a small amount of bias through regularization, both variance and Mean Squared Error (MSE) of prediction can be greatly reduced. Ridge Regression performs coefficient shrinkage but does not set any coefficient exactly equal to zero. LASSO, however, both shrinks coefficients and performs variable selection by forcing some coefficients to become exactly zero — a key advantage in high-dimensional ecological and epidemiological datasets.

6.1.2 Problem Formulation

The LASSO estimator is obtained by minimizing the penalized residual sum of squares:

$$\min_{\beta} \left\{ \sum_{i=1}^n \left(y_i - \sum_{j=1}^p \beta_j x_{ij} \right)^2 + \lambda \sum_{j=1}^p |\beta_j| \right\}$$

An equivalent constrained formulation is:

$$\min \sum_{i=1}^n \left(y_i - \sum_{j=1}^p \beta_j x_{ij} \right)^2 \quad \text{subject to} \quad \sum_{j=1}^p |\beta_j| \leq t$$

where there is a one-to-one correspondence between λ and t . In matrix notation:

$$\min_{\beta} \{ \|Y - X\beta\|_2^2 + \lambda \|\beta\|_1 \}$$

where Y is an $n \times 1$ response vector, X is an $n \times p$ design matrix, β is a $p \times 1$ coefficient vector, and

$$\|\beta\|_1 = \sum_{j=1}^p |\beta_j|$$

is the L_1 -norm penalty. The parameter $\lambda \geq 0$ controls the degree of regularization: larger λ imposes stronger shrinkage and produces sparser models.

6.1.3 LASSO Estimator and Soft Thresholding

The solution to the LASSO optimization problem yields the **soft-thresholding estimator**:

$$\hat{\beta}_j^L = \text{sign}(\hat{\beta}_j^0) \left(\left| \hat{\beta}_j^0 \right| - \frac{\lambda}{2} \right)_+ = \begin{cases} \hat{\beta}_j^0 + \frac{\lambda}{2}, & \hat{\beta}_j^0 < 0 \\ \hat{\beta}_j^0 - \frac{\lambda}{2}, & \hat{\beta}_j^0 > 0 \\ 0, & \text{otherwise} \end{cases}$$

where $\hat{\beta}_j^0$ is the OLS estimator and $(a)_+ = \max(a, 0)$.

This result shows that LASSO uniformly shrinks each OLS coefficient by $\lambda/2$ toward zero. When the OLS coefficient is smaller than the shrinkage threshold, the LASSO estimate becomes exactly zero, removing that predictor from the model entirely. This is the fundamental mechanism through which LASSO achieves simultaneous estimation and variable selection.

The tuning parameter λ is determined from the constraint:

$$\sum_{j=1}^p \left| \hat{\beta}_j^L \right| \leq t$$

In practice, λ is selected via cross-validation to optimize predictive performance on held-out data.

6.1.4 Important Features of LASSO

1. **Shrinkage:** Regression coefficients are shrunk toward zero, reducing variance and model complexity.
2. **Variable Selection:** Coefficients of weak or irrelevant predictors are forced to exactly zero, producing sparse and interpretable models.

3. **Improved Prediction Accuracy:** By controlling overfitting through regularization, LASSO improves generalization performance, especially in high-dimensional settings where p is large relative to n .
4. **Interpretability:** Since only important variables remain active in the model, biological and ecological interpretation is much easier compared to full OLS models.

6.1.5 Algorithm for Finding the LASSO Solution

For fixed $t \geq 0$, the LASSO solution is obtained through the following iterative procedure:

1. **Initialize:** Begin with the OLS estimator $\hat{\beta}^0$ and set the initial active constraint sign vector as

$$\delta_{i_0} = \text{sign}(\hat{\beta}^0) = \left(\text{sign}(\hat{\beta}_1^0), \dots, \text{sign}(\hat{\beta}_p^0) \right)$$

2. **Formulate Constraints:** The constraint $\sum_{j=1}^p |\beta_j| \leq t$ is equivalently written as 2^p linear constraints:

$$\delta'_i \beta \leq t, \quad i = 1, 2, \dots, 2^p$$

3. **Solve:** Solve the constrained least squares problem subject to the currently active constraints.
4. **Check Violations:** Identify any inactive constraint that becomes violated by the current solution.
5. **Update Active Set:** Add violated constraints to the active set and recompute coefficient estimates.

6. **Repeat:** Continue Steps 3–5 until $\sum_{j=1}^p |\beta_j| \leq t$ is satisfied.

The algorithm terminates in finitely many steps since the total number of possible sign combinations is 2^p , ensuring guaranteed convergence.

6.1.6 Procedure of the LASSO Logistic Regression Model

The LASSO Logistic Regression model was developed through a systematic sequence of preprocessing, variable scaling, model training, and validation steps to identify the major ecological and biological predictors associated with human virus infectivity.

Initially, the response variable `homo_infected` was separated from the predictor variables. The predictor matrix consisted of ecological, environmental, taxonomic, and vertebrate host-related variables, while the response variable represented whether a virus was capable of infecting humans.

The dataset was then divided into training and testing subsets using stratified random sampling. A 70 : 30 train-test split was applied, where 70% of the observations were used for model training and 30% were reserved for independent model evaluation. Stratification was applied to preserve the original proportion of infected and non-infected virus observations within both subsets.

Before model fitting, variable scaling was performed using the **RobustScaler** normalization technique. RobustScaler was specifically chosen because the dataset contains

highly skewed ecological variables, extreme outliers, sparse distributions, and zero-inflated host counts. Traditional normalization approaches such as `StandardScaler` are highly sensitive to extreme observations and may produce unstable coefficient estimation under such conditions.

`RobustScaler` reduces the influence of outliers by scaling variables according to the median and interquartile range (IQR) instead of the mean and standard deviation. The transformation is mathematically expressed as:

$$X_{\text{scaled}} = \frac{X - \text{Median}(X)}{\text{IQR}(X)} \quad (6.1)$$

where:

$$\text{IQR}(X) = Q_3 - Q_1$$

This preprocessing approach is particularly suitable for epidemiological and ecological datasets because biological count variables often show heavy-tailed distributions and substantial variance imbalance.

After preprocessing, the LASSO Logistic Regression model was constructed using L1 regularization. The LASSO framework penalizes large regression coefficients and automatically shrinks weak or biologically redundant variables toward zero. Consequently, the model simultaneously performs:

- predictive classification,
- variable selection,
- and dimensionality reduction.

The model was trained using the `liblinear` optimization solver with balanced class weighting to address class imbalance between human-infecting and non-human-infecting viruses.

The regularization strength was controlled using:

$$C = 0.4$$

where smaller values of C impose stronger regularization and greater coefficient shrinkage.

After model fitting, prediction was performed on the independent testing dataset. Model performance was then evaluated using multiple classification metrics including accuracy, precision, recall, F1-score, confusion matrix analysis, cross-validation performance, and ROC-AUC analysis.

Finally, the regression coefficients estimated by the LASSO model were interpreted biologically to identify the most influential ecological predictors associated with zoonotic spillover and human infection risk.

Overall, the LASSO modeling framework provides a robust, interpretable, and biologically meaningful approach for identifying ecological drivers of virus human infectivity while effectively handling multicollinearity, outliers, and sparse high-dimensional data.

6.1.7 Cross-Validation of the LASSO Logistic Regression Model

To evaluate the predictive stability and generalization capability of the LASSO Logistic Regression model, we carried out cross-validation analysis using multiple values of the regularization parameter C .

Cross-validation is a widely used model validation technique that estimates model performance on unseen data by repeatedly partitioning the dataset into multiple training and validation subsets. Compared to a single train-test split, cross-validation reduces sampling bias and provides a more reliable estimate of predictive performance and model robustness.

In LASSO Logistic Regression, the regularization parameter C is defined as:

$$C = \frac{1}{\lambda}$$

where λ is the regularization strength.

Smaller values of C correspond to stronger regularization, forcing the model coefficients to shrink more aggressively toward zero. This process promotes variable selection and reduces model complexity. Conversely, larger values of C decrease the strength of regularization, allowing the model to fit more complex relationships within the data.

Figure 6.1 shows the cross-validation accuracy of the LASSO model across different values of $\log(C)$.

The results show that very small values of C give lower predictive accuracy due to excessive regularization and model underfitting. As the value of C increases, classification accuracy improves greatly, suggesting that moderate regularization enables the model to capture important ecological, virological, and epidemiological relationships more effectively.

The cross-validation accuracy stabilizes at approximately 0.83, showing that the selected LASSO model achieves strong predictive consistency and good generalization performance without severe overfitting. The relatively smooth behavior of the validation curve further shows the robustness and stability of the selected variable set across varying levels of regularization.

Overall, the cross-validation analysis confirms that the LASSO Logistic Regression framework provides reliable classification performance for distinguishing human-infecting viruses from non-human-associated viral pathogens while maintaining model interpretability and variable selection capability.

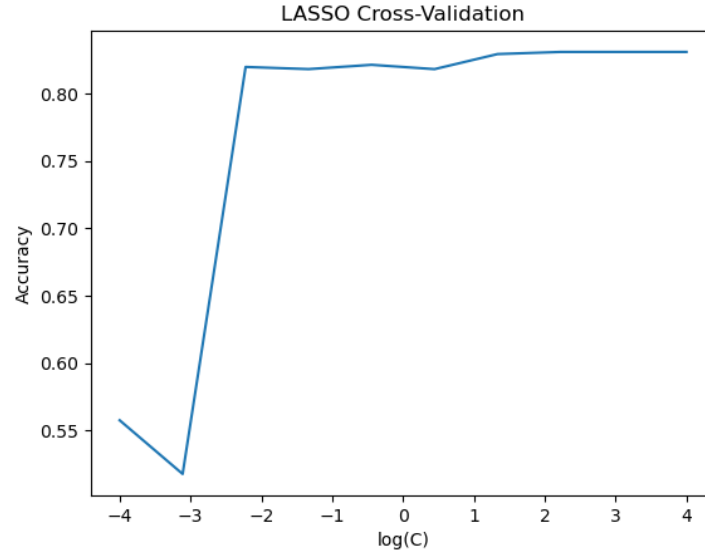


Figure 6.1: Cross-Validation Accuracy of the LASSO Logistic Regression Model Across Different Values of $\log(C)$

6.1.8 Interpretation of LASSO Regression Coefficients

The coefficients obtained from the LASSO Logistic Regression model were analyzed to identify the most important predictors associated with human infectivity status (`homo_infected`). LASSO regularization performs automatic variable selection by shrinking less important coefficients toward zero, thereby improving model interpretability and reducing dimensional complexity.

Among all variables, `Virus_Group` showed the largest coefficient magnitude (-2.733), indicating that virus ecological classification is one of the strongest determinants of zoonotic transmission potential. Host-related ecological variables such as `Primates`, `Diprotodontia`, `Chiroptera`, `Carnivora`, and `Rodentia` showed positive coefficients, suggesting that mammalian reservoirs play a major role in viral spillover and human infectivity.

The variable `vector_C` also showed a relatively large negative coefficient, highlighting the importance of vector classification in virus transmission dynamics. Similarly, `baltimore` showed a negative coefficient, indicating that virus genomic classification contributes to infection prediction.

Notably, variables such as `Lagomorpha` and `climate_group` obtained coefficients equal to zero, showing that these predictors added little to the final classification model and were effectively removed by the LASSO regularization process.

Overall, the LASSO coefficient analysis suggests that virus ecological classification, mammalian host diversity, and vector-related characteristics are among the most important determinants associated with zoonotic spillover and viral pathogenicity.

Table 6.1: Important LASSO Regression Coefficients

Variable	Coefficient
Virus_Group	-2.732772
Primates	1.634704
Diprotodontia	0.892766
Eulipotyphla	0.536986
vector_C	-0.489091
Chiroptera	0.484684
Carnivora	0.433587
Rodentia	0.423046
baltimore	-0.359701
Pilosa	-0.253077
Aves	0.095431
Didelphimorphia	0.074438
Artiodactyla	-0.016815
continent	-0.001144
Lagomorpha	0.000000
climate_group	0.000000

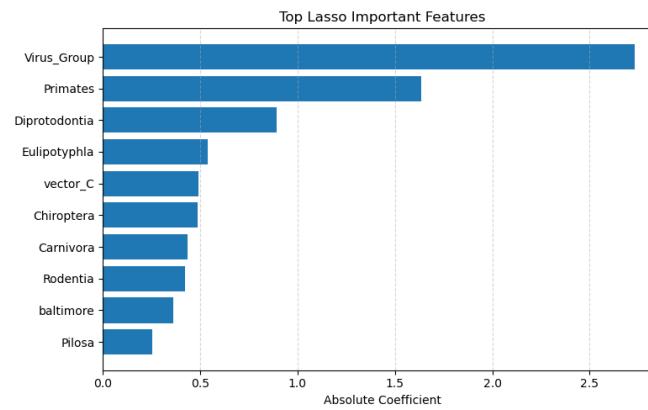


Figure 6.2: Variable Importance Based on LASSO Regression Coefficients

Chapter 7

Ridge Regression

7.1 Ridge Regression

Introduction

Ridge Regression, introduced by Hoerl and Kennard (1970), is a regularization technique that addresses multicollinearity and overfitting in regression analysis by imposing an L_2 penalty on the regression coefficients. Unlike OLS, which minimizes only the residual sum of squares, Ridge regression shrinks all coefficients toward zero simultaneously — without eliminating any predictor entirely. This makes it particularly useful when predictors are highly correlated and coefficient estimation is unstable.

Consider the regression model

$$V = Z\beta + \varepsilon, \quad E(\varepsilon) = 0$$

where V is an $n \times 1$ response vector, Z is an $n \times p$ design matrix, and β is a $p \times 1$ coefficient vector.

Problem Formulation

The ridge estimator minimizes the penalized residual sum of squares:

$$\min_{\beta} \left\{ \sum_{i=1}^n (V_i - Z'_i \beta)^2 + k \sum_{j=1}^p \beta_j^2 \right\}$$

where $k > 0$ is the tuning parameter controlling shrinkage strength. Larger values of k produce greater shrinkage. An equivalent constrained formulation is:

$$\min_{\beta} \sum_{i=1}^n (V_i - Z'_i \beta)^2 \quad \text{subject to} \quad \sum_{j=1}^p \beta_j^2 \leq t$$

There is a one-to-one correspondence between k and t .

Ridge Estimator

Minimizing the penalized objective function with respect to β yields:

$$\hat{\beta}_k = (Z'Z + kI_p)^{-1} Z'V$$

where I_p is the $p \times p$ identity matrix. This can equivalently be written as:

$$\hat{\beta}_k = [I + k(Z'Z)^{-1}]^{-1} \hat{\beta} = M\hat{\beta}$$

where $M = [I + k(Z'Z)^{-1}]^{-1}$ and $\hat{\beta} = \hat{\beta}_{OLS}$.

A small quantity k is added to the diagonal elements of $Z'Z$, which may be viewed as a *ridge* on the diagonal — hence the name **ridge regression**.

Bias of the Ridge Estimator

Taking the expectation of $\hat{\beta}_k$:

$$E(\hat{\beta}_k) = (Z'Z + kI_p)^{-1} Z'Z\beta$$

Therefore,

$$\text{Bias}(\hat{\beta}_k) = [(Z'Z + kI_p)^{-1} Z'Z - I] \beta \neq 0$$

The ridge estimator is a **biased estimator**. However, this bias is introduced intentionally to achieve a reduction in variance.

Mean Squared Error of the Ridge Estimator

The MSE of $\hat{\beta}_k$ decomposes into variance and squared bias:

$$\text{MSE}(\hat{\beta}_k) = E[(\hat{\beta}_k - \beta)'(\hat{\beta}_k - \beta)]$$

Let $W = (Z'Z + kI_p)^{-1}$. Using the decomposition $\hat{\beta}_k = M\hat{\beta}$ and $E[(\hat{\beta} - \beta)(\hat{\beta} - \beta)'] = \sigma^2(Z'Z)^{-1}$:

$$\text{MSE}(\hat{\beta}_k) = \sigma^2 \text{tr}(W) - k\sigma^2 \text{tr}(W^2) + k^2 \beta' W^2 \beta$$

The first two terms represent the **variance component** and the third term represents the **squared bias component**.

Eigenvalue Representation

Let $\lambda_1, \lambda_2, \dots, \lambda_p$ be the eigenvalues of $Z'Z$. Then the eigenvalues of $W = (Z'Z + kI_p)^{-1}$ are $\frac{1}{\lambda_i + k}$, so that:

$$\text{tr}(W) = \sum_{i=1}^p \frac{1}{\lambda_i + k}, \quad \text{tr}(W^2) = \sum_{i=1}^p \frac{1}{(\lambda_i + k)^2}$$

Defining $\alpha = P\beta$ where $Z'Z = P'\Delta P$:

$$\text{MSE}(\hat{\beta}_k) = \sum_{i=1}^p \frac{\lambda_i}{(\lambda_i + k)^2} + \sum_{i=1}^p \frac{\alpha_i^2}{(1 + \frac{\lambda_i}{k})^2}$$

Key result: When $k = 0$, $\text{MSE}(\hat{\beta}_k) = \sum_{i=1}^p \frac{1}{\lambda_i} = \text{MSE}(\hat{\beta})$, the OLS MSE. For $0 < k < 1$:

$$\text{MSE}(\hat{\beta}_k) < \text{MSE}(\hat{\beta})$$

Thus, ridge regression produces a smaller MSE than OLS by trading a small increase in bias for a larger reduction in variance. As k increases, ridge estimates shrink toward zero but never become exactly zero — a key distinction from LASSO.

Ridge Trace

In practice, the ridge parameter k is selected by computing $\hat{\beta}_1(k), \hat{\beta}_2(k), \dots, \hat{\beta}_p(k)$ for values of $k \in [0, 1]$ and plotting against k . The resulting graph is called the **ridge trace**, and an appropriate k is chosen at the point where the coefficients stabilize.

Important Features of Ridge Regression

1. **Shrinkage without elimination:** All coefficients are shrunk toward zero but none become exactly zero, retaining all predictors.
2. **Multicollinearity control:** Adding kI_p to $Z'Z$ ensures invertibility even under near-perfect collinearity.
3. **Bias-variance tradeoff:** Ridge reduces variance substantially at the cost of a small controlled bias, improving overall MSE.
4. **Stability:** Coefficient estimates are more stable than OLS under high-dimensional or collinear settings.

7.1.1 Procedure of the Ridge Logistic Regression Model

The Ridge Logistic Regression model was developed through a systematic sequence of preprocessing, feature scaling, model training, and validation steps to identify the major ecological and biological predictors associated with human viral infectivity.

Initially, the response variable `homo_infected` was separated from the predictor variables. The predictor matrix consisted of ecological, environmental, taxonomic, and vertebrate host-related variables, while the response variable represented whether a virus was capable of infecting humans.

The dataset was subsequently divided into training and testing subsets using stratified random sampling. A 70 : 30 train-test split was applied, where 70% of the observations were used for model training and 30% were reserved for independent model evaluation. Stratification was applied to preserve the original proportion of infected and non-infected viral observations within both subsets.

Before model fitting, feature scaling was performed using the **RobustScaler** normalization technique. RobustScaler was specifically selected because the dataset contains highly skewed ecological variables, extreme outliers, sparse distributions, and zero-inflated host counts. Traditional normalization approaches such as StandardScaler are highly sensitive to extreme observations and may produce unstable coefficient estimation under such conditions.

RobustScaler reduces the influence of outliers by scaling variables according to the median and interquartile range (IQR) instead of the mean and standard deviation. The transformation is mathematically expressed as:

$$X_{\text{scaled}} = \frac{X - \text{Median}(X)}{\text{IQR}(X)} \quad (7.1)$$

where:

$$\text{IQR}(X) = Q_3 - Q_1$$

This preprocessing approach is particularly suitable for epidemiological and ecological datasets because biological count variables often exhibit heavy-tailed distributions and substantial variance imbalance.

After preprocessing, the Ridge Logistic Regression model was constructed using L2 regularization. Ridge regularization penalizes large regression coefficients while retaining all predictor variables within the model. This approach helps reduce multicollinearity, stabilize coefficient estimation, and improve model generalization performance without eliminating biologically relevant variables.

Consequently, the Ridge Logistic Regression framework simultaneously provides:

- predictive classification,
- coefficient stabilization,
- and multicollinearity reduction.

The model was trained using the `liblinear` optimization solver with balanced class weighting to address class imbalance between human-infecting and non-human-infecting viruses.

The regularization strength was controlled using:

$$C = 0.4$$

where smaller values of C impose stronger regularization and greater coefficient shrinkage.

After model fitting, prediction was performed on the independent testing dataset. Model performance was subsequently evaluated using multiple classification metrics including accuracy, precision, recall, F1-score, confusion matrix analysis, cross-validation performance, and ROC-AUC analysis.

Finally, the regression coefficients estimated by the Ridge Logistic Regression model were interpreted biologically to identify the most influential ecological predictors associated with zoonotic spillover and human infection risk.

Overall, the Ridge Logistic Regression framework provides a robust, stable, and biologically interpretable approach for identifying ecological drivers of viral human infectivity while effectively handling multicollinearity, outliers, and high-dimensional ecological data.

7.1.2 Cross-Validation of the Ridge Logistic Regression Model

To evaluate the predictive stability and generalization capability of the Ridge Logistic Regression model, cross-validation analysis was conducted using multiple values of the regularization parameter C .

Figure 7.1 illustrates the cross-validation accuracy of the Ridge model across different values of $\log(C)$. The results demonstrate that very small values of C , corresponding to strong regularization, produced comparatively lower predictive accuracy due to excessive coefficient shrinkage and model underfitting.

As the value of C increased, model performance improved substantially, indicating that moderate regularization allowed the model to capture important ecological and epidemiological relationships more effectively. The cross-validation accuracy gradually stabilized between 0.86 and 0.87, suggesting strong predictive consistency and stable model generalization performance.

The relatively smooth behavior of the validation curve indicates that the Ridge Logistic Regression model remained robust across varying levels of regularization strength, reflecting stable coefficient estimation and reduced multicollinearity effects.

Overall, the cross-validation analysis confirms that the Ridge Logistic Regression framework provides reliable classification performance for distinguishing human-infecting viruses from non-human-associated viral pathogens while maintaining model stability and interpretability.

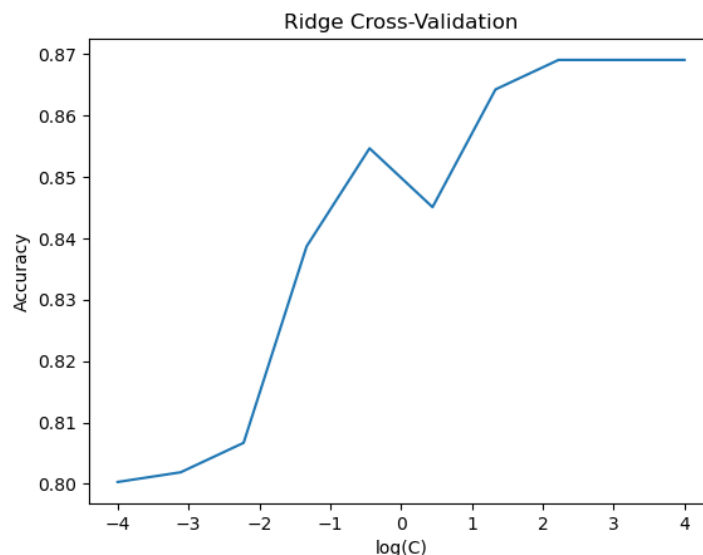


Figure 7.1: Cross-Validation Accuracy of the Ridge Logistic Regression Model Across Different Values of $\log(C)$

7.1.3 Ridge Logistic Regression Coefficient Interpretation

The coefficients obtained from the Ridge Logistic Regression model were analyzed to evaluate the contribution and directional influence of predictor variables on human infectivity status (`homo_infected`). Ridge regularization was applied to reduce multicollinearity and improve model stability while retaining all predictor variables within the model.

Among all variables, `Virus_Group` exhibited the largest coefficient magnitude (-2.504), indicating that viral ecological classification strongly influences the probability of human infection. Host-related ecological variables such as `Primates`, `Eulipotyphla`, `Diprotodontia`, `Rodentia`, and `Chiroptera` showed positive coefficients, suggesting that mammalian host diversity substantially contributes to zoonotic spillover and viral pathogenicity.

The variable `vector_C` demonstrated a negative coefficient, indicating that vector classification also affects viral transmission potential. Environmental and geographical variables such as `climate_group` and `continent` displayed relatively smaller coefficient magnitudes, suggesting comparatively weaker influence on human infectivity prediction.

Overall, the Ridge Logistic Regression analysis suggests that viral ecological classification and mammalian reservoir diversity are among the strongest determinants associated with zoonotic transmission and human infection risk.

Table 7.1: Ridge Logistic Regression Coefficients

Feature	Coefficient
Virus_Group	-2.504439
Primates	2.124624
Eulipotyphla	1.100861
Diprotodontia	1.100807
Didelphimorphia	0.782863
Rodentia	0.672105
vector_C	-0.656093
Chiroptera	0.602345
Carnivora	0.456248
baltimore	-0.332797
Pilosa	-0.331110
Aves	0.188485
climate_group	-0.164149
continent	-0.079036
Artiodactyla	-0.032749
Lagomorpha	0.009715

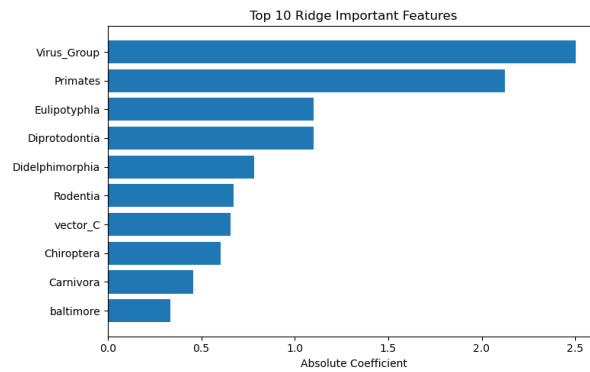


Figure 7.2: Feature Importance Based on Ridge Logistic Regression Coefficients

Chapter 8

Random Forest

8.1 Random Forest Classifier

8.1.1 Introduction

Random Forest is an ensemble machine learning algorithm belonging to the supervised learning category. It combines multiple decision trees built on different subsets of the dataset to improve predictive performance and reduce overfitting. The algorithm is based on *ensemble learning*, where multiple classifiers are combined to solve complex problems.

A random forest consists of many decision trees, where a greater number of trees generally leads to higher accuracy and reduced variance.

8.1.2 Algorithm Steps

The Random Forest algorithm follows these steps:

1. For each tree, select n random records and m random features from the original dataset.
2. Construct individual decision trees for each sample.
3. Each decision tree generates an output.
4. The final output is determined by:
 - **Majority voting** for classification problems
 - **Averaging** for regression problems

8.1.3 Important Features

- **Diversity:** Each tree uses different subsets of variables and data, making trees independent.
- **Immune to Curse of Dimensionality:** Since each tree uses a subset of features, the effective feature space is reduced.
- **Parallelization:** Trees are built independently, allowing efficient parallel computation.

- **Stability:** Final results are based on majority voting, reducing variance.
- **Out-of-Bag (OOB) Evaluation:** Approximately one-third of data is not used for training and serves as a built-in validation set.

8.1.4 Procedure of the Random Forest Classification Model

The Random Forest classification model was developed through systematic data preparation, model training, and performance evaluation steps.

Initially, the response variable `homo_infected` was separated from the predictor variables:

$$X = \text{Predictor Variables}, \quad y = \text{homo_infected}$$

An 80 : 20 stratified train-test split was applied to preserve the proportional distribution of infected and non-infected virus observations across both subsets.

Random Forest does not require variable scaling because decision trees perform data partitioning based on threshold splitting rather than numerical magnitude. This is particularly advantageous for ecological datasets, which often contain skewed distributions, extreme outliers, and sparse observations.

The model was configured with the following hyperparameters:

- **n_estimators = 200:** Number of decision trees in the forest.
- **max_depth = None:** No explicit depth restriction.
- **min_samples_split = 5:** Minimum observations required to split a node.
- **min_samples_leaf = 2:** Minimum observations per leaf node.
- **class_weight = 'balanced':** Compensates for class imbalance.
- **random_state = 42:** Ensures reproducibility.
- **n_jobs = -1:** Uses all available processor cores.

During training, multiple bootstrap samples were generated, and each decision tree was independently trained using random subsets of observations and variables. Final predictions were determined through majority voting.

8.1.5 Classification Performance

The Random Forest model achieved:

$$\text{Accuracy} = 93.85\%, \quad \text{AUC} = 0.9865$$

Confusion Matrix

$$\begin{bmatrix} 76 & 3 \\ 8 & 92 \end{bmatrix}$$

where TN = 76, FP = 3, FN = 8, TP = 92.

Performance Metrics

$$\text{Precision} = \frac{92}{92 + 3} = 0.968, \quad \text{Recall} = \frac{92}{92 + 8} = 0.92$$

$$F_1 = 2 \times \frac{0.968 \times 0.92}{0.968 + 0.92} \approx 0.94$$

Macro and weighted averages: Precision = 0.94, Recall = 0.94, F1-score = 0.94.

8.1.6 Variable Importance Analysis

Variable importance was estimated based on the average reduction in node impurity across all trees. Figure 8.1 and Table 8.1 show the variable importance scores.

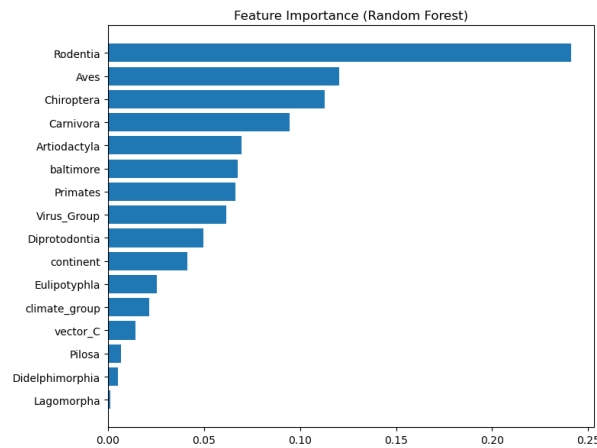


Figure 8.1: Variable Importance of the Random Forest Classifier

Table 8.1: Variable Importance Scores — Random Forest Classifier

Variable	Importance Score
Rodentia	Highest
Aves	High
Chiroptera	High
Carnivora	Moderate
Virus_Group	Moderate
baltimore	Moderate
continent	Moderate
climate_group	Moderate
Artiodactyla	Moderate
Diprotodontia	Moderate
Eulipotyphla	Moderate
Primates	Moderate
Pilosa	Low
Didelphimorphia	Low
Lagomorpha	Low
vector_C	Low

Rodentia was the most influential predictor, consistent with rodents being major reservoirs of zoonotic viruses. **Aves** and **Chiroptera** also showed strong importance, reflecting the role of birds in arbovirus amplification and bats as reservoirs for emerging viruses.

The moderate importance of **Virus_Group**, **baltimore**, **continent**, and **climate_group** suggests that viral classification, geography, and climate collectively influence transmission dynamics.

Overall, mammalian and avian reservoir ecology dominate the biological signal for predicting viral human infectivity.

8.1.7 ROC Curve Analysis

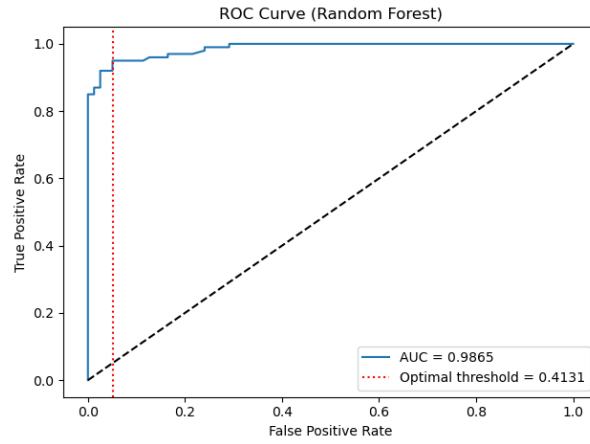


Figure 8.2: ROC Curve of the Random Forest Classifier

The ROC curve plots True Positive Rate (Sensitivity) against False Positive Rate (1 – Specificity):

$$TPR = \frac{TP}{TP + FN}, \quad FPR = \frac{FP}{FP + TN}$$

The model achieved $AUC = 0.9865$, indicating outstanding discriminatory capability. The curve rises sharply toward the upper-left corner, confirming high sensitivity and specificity. The optimal classification threshold was approximately 0.4131.

8.1.8 Train vs Test MSE and RMSE

MSE (Brier Score) and RMSE were computed:

$$MSE = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{p}_i)^2, \quad RMSE = \sqrt{MSE}$$

Table 8.2: Train vs Test MSE and RMSE — Random Forest Classifier

Metric	Train	Test
MSE	0.0460	0.0525
RMSE	0.2146	0.2291

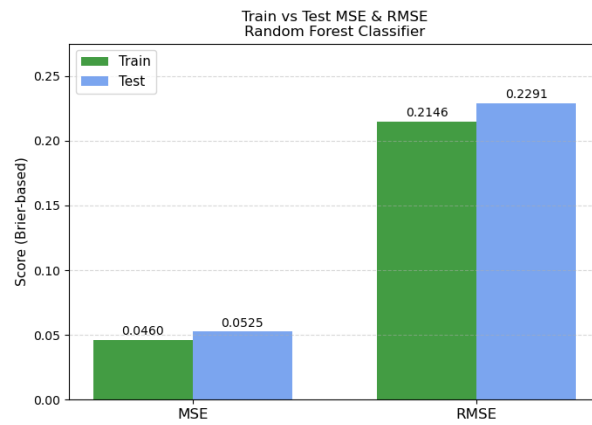


Figure 8.3: Train vs Test MSE and RMSE — Random Forest Classifier

The small gap between training and testing errors confirms good generalization without significant overfitting.

Chapter 9

Support Vector Machine

9.1 Support Vector Machine (SVM)

9.1.1 Introduction

Support Vector Machine (SVM) is a supervised learning algorithm developed by Vapnik and Cortes (1995), based on Statistical Learning Theory and Structural Risk Minimization. SVM constructs an optimal hyperplane in a high-dimensional feature space that maximally separates observations from different classes.

Unlike probabilistic classifiers, SVM directly optimizes the geometric margin between classes, making it particularly robust for small-sample, high-dimensional, and nonlinear classification problems. The algorithm is well-suited for ecological and epidemiological datasets where the number of predictors may exceed observations ($p \gg n$).

9.1.2 Basic Concept

Given training data $D = \{(x_i, y_i)\}_{i=1}^n$ with $x_i \in \mathbb{R}^p$ and $y_i \in \{-1, +1\}$, SVM seeks a separating hyperplane:

$$w^T x + b = 0$$

where w is the weight vector and b is the bias term. The decision rule is:

$$f(x) = \text{sign}(w^T x + b)$$

Among all possible separating hyperplanes, the optimal one maximizes the margin — the perpendicular distance between the hyperplane and the nearest training observations (support vectors) from each class.

9.1.3 Optimization Problem

Hard Margin SVM (Linearly Separable)

For perfectly separable data, the constraints are:

$$y_i(w^T x_i + b) \geq 1, \quad \forall i$$

The primal optimization problem is:

$$\min_{w, b} \frac{1}{2} \|w\|^2 \quad \text{subject to} \quad y_i(w^T x_i + b) \geq 1$$

Soft Margin SVM (Non-Linearly Separable)

Real-world data are rarely perfectly separable. Slack variables $\xi_i \geq 0$ are introduced:

$$y_i(w^T x_i + b) \geq 1 - \xi_i, \quad \xi_i \geq 0$$

The modified optimization is:

$$\min_{w,b,\xi} \frac{1}{2} \|w\|^2 + C \sum_{i=1}^n \xi_i$$

where $C > 0$ controls the trade-off between margin width and misclassification penalty.

9.1.4 Dual Formulation and Kernel Trick

Using Lagrange multipliers $\alpha_i \geq 0$, the dual problem is:

$$\max_{\alpha} \sum_{i=1}^n \alpha_i - \frac{1}{2} \sum_{i,j} \alpha_i \alpha_j y_i y_j K(x_i, x_j)$$

subject to:

$$\alpha_i \geq 0, \quad \sum_{i=1}^n \alpha_i y_i = 0$$

The kernel function $K(x_i, x_j) = \phi(x_i)^T \phi(x_j)$ enables nonlinear classification without explicit feature transformation.

9.1.5 Kernel Functions

Three kernel functions were evaluated:

Kernel	Formula
Linear	$K(x_i, x_j) = x_i^T x_j$
RBF	$K(x_i, x_j) = \exp(-\gamma \ x_i - x_j\ ^2)$
Polynomial	$K(x_i, x_j) = (\gamma x_i^T x_j + 1)^3$

Table 9.1: SVM Kernel Functions

The RBF kernel is the most widely used, mapping observations into an infinite-dimensional feature space. The Linear kernel is computationally efficient and effective when strong linear separability exists. The Polynomial kernel captures feature interactions but may overfit.

9.1.6 Procedure of the SVM Model

The response variable `homo_infected` was separated from predictors: $X = \text{Predictors}$, $y = \text{homo_infected}$. An 80 : 20 stratified train-test split was applied (Train: 716, Test: 179).

SVM is sensitive to feature scale, so `StandardScaler` was applied:

$$X_{\text{scaled}} = \frac{X - \mu}{\sigma}$$

Class imbalance was addressed using `class_weight = 'balanced'`. Three kernels were evaluated with $C = 1.0$. The RBF kernel was selected for hyperparameter tuning via `GridSearchCV` with 5-fold cross-validation:

$$C \in \{0.1, 1, 10\}, \quad \gamma \in \{\text{scale}, 0.1, 0.01\}$$

9.1.7 Kernel Comparison and Performance Summary

Kernel	Train Acc	Test Acc	Train MSE	Test MSE	Train RMSE	Test RMSE
RBF	0.8813	0.8268	0.1187	0.1732	0.3446	0.4162
Linear	0.8492	0.8268	0.1508	0.1732	0.3884	0.4162
Polynomial	0.7821	0.7542	0.2179	0.2458	0.4668	0.4958

Table 9.2: Performance Summary Across SVM Kernels

The RBF and Linear kernels achieved identical test accuracy (82.68%), with Linear showing superior AUC (0.9241). The Polynomial kernel performed weakest across all metrics.

9.1.8 Confusion Matrix Analysis

RBF Kernel

$$\begin{pmatrix} 70 & 9 \\ 22 & 78 \end{pmatrix} \quad (\text{TN}=70, \text{FP}=9, \text{FN}=22, \text{TP}=78)$$

Linear Kernel

$$\begin{pmatrix} 77 & 2 \\ 29 & 71 \end{pmatrix} \quad (\text{TN}=77, \text{FP}=2, \text{FN}=29, \text{TP}=71)$$

Polynomial Kernel

$$\begin{pmatrix} 78 & 1 \\ 43 & 57 \end{pmatrix} \quad (\text{TN}=78, \text{FP}=1, \text{FN}=43, \text{TP}=57)$$

The Linear kernel produced the fewest False Positives ($\text{FP} = 2$). The Polynomial kernel produced the most False Negatives ($\text{FN} = 43$), which is critically undesirable from an epidemiological surveillance perspective. The RBF kernel offered the best balance between FP and FN rates.

9.1.9 Classification Performance

RBF Kernel

$$\text{Accuracy} = 0.8268, \quad \text{AUC} = 0.9160$$

Class	Precision	Recall	F1-score	Support
0 (Non-infected)	0.76	0.89	0.82	79
1 (Infected)	0.90	0.78	0.84	100
Macro Avg	0.83	0.83	0.83	179
Weighted Avg	0.84	0.83	0.83	179

Linear Kernel

$$\text{Accuracy} = 0.8268, \quad \text{AUC} = 0.9241$$

Class	Precision	Recall	F1-score	Support
0 (Non-infected)	0.73	0.97	0.83	79
1 (Infected)	0.97	0.71	0.82	100
Macro Avg	0.85	0.84	0.83	179
Weighted Avg	0.86	0.83	0.83	179

The Linear kernel showed high precision (0.97) for infected viruses, meaning viruses predicted as human-infecting are likely genuine zoonotic threats. However, lower recall (0.71) indicates some human-infecting viruses were missed.

Polynomial Kernel

$$\text{Accuracy} = 0.7542, \quad \text{AUC} = 0.9016$$

Class	Precision	Recall	F1-score	Support
0 (Non-infected)	0.64	0.99	0.78	79
1 (Infected)	0.98	0.57	0.72	100
Macro Avg	0.81	0.78	0.75	179
Weighted Avg	0.83	0.75	0.75	179

The Polynomial kernel exhibited the lowest recall (0.57) for infected viruses, meaning 43% of zoonotic pathogens were misclassified — the most serious failure from an epidemiological perspective.

9.1.10 Support Vector Count

Kernel	Class 0 SVs	Class 1 SVs	Total SVs
RBF	123	158	281
Linear	116	137	253
Polynomial	188	239	427

Table 9.3: Number of Support Vectors per Kernel

The Polynomial kernel required the most support vectors (427), reflecting a complex decision boundary. The Linear kernel required the fewest (253), consistent with its simpler structure. Fewer support vectors generally indicate better generalization.

9.1.11 Hyperparameter Tuning Results

GridSearchCV with 5-fold cross-validation identified optimal parameters for RBF kernel:

$$C = 10, \quad \gamma = \text{scale}$$

Best cross-validation accuracy: 0.8743.

9.1.12 Tuned Model Performance

Metric	Train	Test
Accuracy	0.8883	0.8212
MSE	0.1117	0.1788
RMSE	0.3343	0.4228
AUC	0.9282	

Table 9.4: Tuned SVM Performance ($C = 10$, RBF)

The tuned model improved AUC to 0.9282 compared to default RBF (0.9160). However, the train-test accuracy gap ($\Delta = 0.0671$) indicates mild overfitting due to aggressive $C = 10$.

Class	Precision	Recall	F1-score	Support
0 (Non-infected)	0.76	0.87	0.81	79
1 (Infected)	0.89	0.80	0.84	100
Macro Avg	0.82	0.83	0.82	179
Weighted Avg	0.83	0.82	0.82	179

9.1.13 ROC Curve Analysis

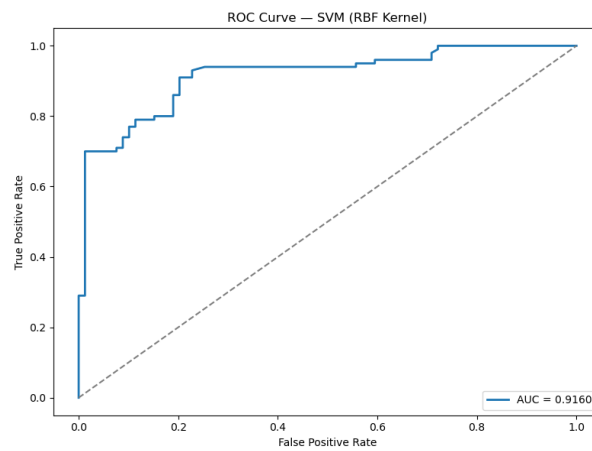


Figure 9.1: ROC Curve — RBF Kernel SVM

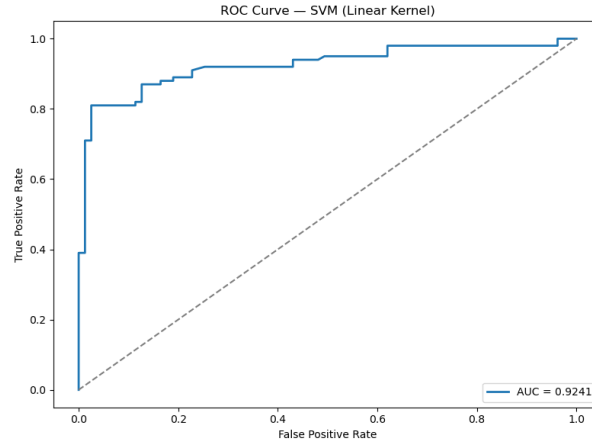


Figure 9.2: ROC Curve — Linear Kernel SVM

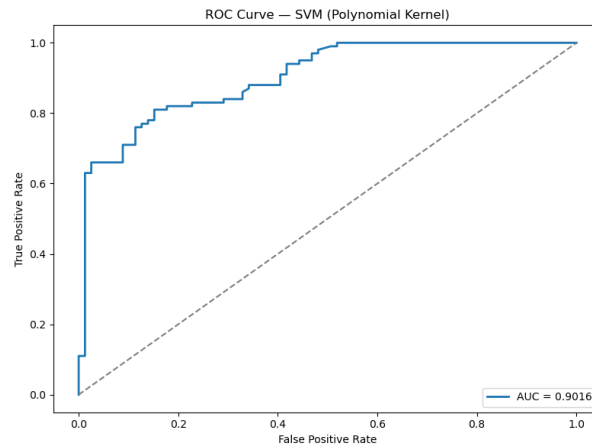


Figure 9.3: ROC Curve — Polynomial Kernel SVM

The ROC curves confirm that RBF and Linear kernels achieved strong discriminatory capability, while the Polynomial kernel showed weaker performance.

9.1.14 Train vs Test MSE and RMSE

Kernel	Train MSE	Test MSE	Train RMSE	Test RMSE
RBF	0.1187	0.1732	0.3446	0.4162
Linear	0.1508	0.1732	0.3884	0.4162
Polynomial	0.2179	0.2458	0.4668	0.4958
Tuned RBF	0.1117	0.1788	0.3343	0.4228

Table 9.5: Train vs Test MSE and RMSE Across SVM Kernels

The RBF and Linear kernels achieved identical Test MSE (0.1732). The Polynomial kernel produced the highest Test MSE (0.2458). The tuned model showed slightly higher Test MSE (0.1788) but improved AUC.

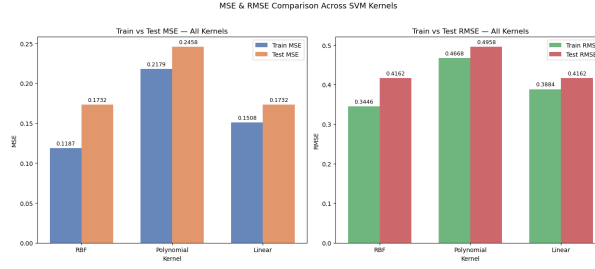


Figure 9.4: Train vs Test MSE and RMSE Across All SVM Kernels

9.1.15 Conclusion

SVM demonstrated strong predictive performance for viral human infectivity classification, achieving best test accuracy of 82.68% and ROC-AUC of 0.9241 (Linear kernel) and 0.9282 (tuned RBF). Among the three kernels, RBF and Linear kernels outperformed Polynomial across all metrics. Hyperparameter tuning improved AUC but introduced mild overfitting.

Compared with ensemble methods (Random Forest: 93.85%, XGBoost: 94.97%), SVM showed lower predictive accuracy, reflecting the inherent limitation of single-kernel classifiers in modeling the complex, non-linear ecological interaction patterns associated with zoonotic spillover. Nevertheless, SVM's strong theoretical foundation and kernel flexibility make it a valuable component of the comparative machine learning framework.

Chapter 10

XGBoost Classifier

10.1 XGBoost Classifier

10.1.1 Introduction

XGBoost (Extreme Gradient Boosting) is a supervised machine learning algorithm based on Gradient Boosting Decision Trees (GBDT). It combines multiple weak learners (decision trees) sequentially to produce a strong predictive model. XGBoost improves performance through boosting, regularization, and second-order optimization, making it highly effective for classification problems.

For binary classification, XGBoost predicts probabilities using the logistic function and minimizes logistic loss.

10.1.2 Mathematical Formulation

Given a training dataset $D = \{(x_i, y_i)\}_{i=1}^n$ with $x_i \in \mathbb{R}^p$ and $y_i \in \{0, 1\}$, the prediction is:

$$\hat{y}_i = \sum_{k=1}^K f_k(x_i)$$

where f_k is the k^{th} decision tree and K is the total number of trees.

The objective function minimized by XGBoost is:

$$\text{Obj} = \sum_{i=1}^n l(y_i, \hat{y}_i) + \sum_{k=1}^K \Omega(f_k)$$

where $l(y_i, \hat{y}_i)$ is the loss function and $\Omega(f_k)$ is the regularization term:

$$\Omega(f) = \gamma T + \frac{1}{2} \lambda \sum_{j=1}^T w_j^2$$

Here, T is the number of leaves, w_j are leaf weights, and γ, λ are regularization parameters.

10.1.3 Key Results

Using second-order Taylor expansion, the objective at iteration t is:

$$L^{(t)} \approx \sum_{i=1}^n \left[g_i f_t(x_i) + \frac{1}{2} h_i f_t^2(x_i) \right] + \Omega(f_t)$$

where $g_i = \partial l / \partial \hat{y}_i$ (gradient) and $h_i = \partial^2 l / \partial \hat{y}_i^2$ (Hessian).

For logistic loss:

$$g_i = p_i - y_i, \quad h_i = p_i(1 - p_i), \quad p_i = \frac{1}{1 + e^{-\hat{y}_i}}$$

The optimal leaf weight for node j is:

$$w_j^* = -\frac{G_j}{H_j + \lambda}, \quad \text{where } G_j = \sum g_i, \quad H_j = \sum h_i$$

The gain from splitting is:

$$\text{Gain} = \frac{1}{2} \left[\frac{G_L^2}{H_L + \lambda} + \frac{G_R^2}{H_R + \lambda} - \frac{(G_L + G_R)^2}{H_L + H_R + \lambda} \right] - \gamma$$

A split is accepted if $\text{Gain} > 0$. The model update with learning rate η is:

$$\hat{y}^{(t)} = \hat{y}^{(t-1)} + \eta f_t(x), \quad 0 < \eta < 1$$

10.1.4 Algorithm

Algorithm 1 XGBoost Classification Algorithm

- 1: Input: $D = \{(x_i, y_i)\}_{i=1}^n$, K = number of trees
 - 2: Initialize $\hat{y}_i^{(0)} = 0$
 - 3: **for** $t = 1$ to K **do**
 - 4: Compute gradients g_i and Hessians h_i
 - 5: Construct decision tree using g_i and h_i
 - 6: Compute split gain and select best split
 - 7: Compute optimal leaf weights w_j^*
 - 8: Update: $\hat{y}^{(t)} = \hat{y}^{(t-1)} + \eta f_t(x)$
 - 9: **end for**
 - 10: Compute probability: $P(y = 1|x) = \frac{1}{1 + e^{-\hat{y}}}$
 - 11: Assign: $\hat{y} = \begin{cases} 1, & P > 0.5 \\ 0, & \text{otherwise} \end{cases}$
-

10.1.5 Advantages and Limitations

Advantages: High predictive accuracy, handles nonlinear relationships, reduces overfitting through regularization, efficient for large datasets, handles missing values automatically, provides variable importance, and works well with high-dimensional data.

Limitations: Hyperparameter tuning can be difficult, computationally expensive for deep trees, less interpretable than simple models, and sensitive to noisy datasets.

10.1.6 Predictor Variables Used

After multicollinearity assessment and variable selection, the final predictor set for the XGBoost model included:

- Virus Group, vector_C, baltimore
- continent, climate_group
- Aves, Carnivora, Rodentia, Chiroptera, Primates
- Artiodactyla, Didelphimorphia, Eulipotyphla
- Lagomorpha, Pilosa, Diprotodontia

The response variable was `homo_infected`, indicating whether a virus infects humans.

10.1.7 Model Procedure

The response variable was separated from predictors: $X = \text{Predictors}$, $y = \text{homo_infected}$. An 80 : 20 stratified train-test split was applied. XGBoost is tree-based and does not require variable scaling. Class imbalance was addressed using:

$$\text{scale_pos_weight} = \frac{|y = 0|}{|y = 1|}$$

Key hyperparameters used:

- **n_estimators**: Number of boosting rounds
- **learning_rate** (η): Controls each tree's contribution
- **max_depth**: Controls tree complexity
- γ, λ : Regularization parameters
- **scale_pos_weight**: Compensates for class imbalance
- **random_state = 42**: Reproducibility

10.1.8 Classification Performance

The XGBoost model achieved:

$$\text{Accuracy} = 94.97\%, \quad \text{AUC} = 0.99$$

Confusion Matrix

$$\begin{pmatrix} 76 & 3 \\ 6 & 94 \end{pmatrix}$$

where TN = 76, FP = 3, FN = 6, TP = 94.

Performance Metrics

$$\begin{aligned} \text{Precision} &= \frac{94}{94 + 3} = 0.969, & \text{Recall} &= \frac{94}{94 + 6} = 0.94 \\ F_1 &= 2 \times \frac{0.969 \times 0.94}{0.969 + 0.94} \approx 0.954 \end{aligned}$$

Macro and weighted averages: Precision = 0.95, Recall = 0.95, F1-score = 0.95.

XGBoost produced fewer false negatives (FN = 6) than Random Forest (FN = 8), which is critically important for epidemiological surveillance.

10.1.9 Variable Importance Analysis

Variable importance was assessed using average gain contributed by each variable across all boosting rounds. Figure 10.1 and Table 10.1 show the top 10 most important predictors.

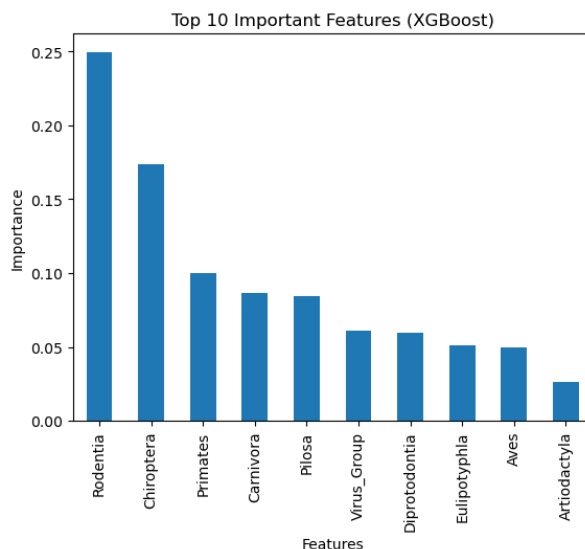


Figure 10.1: Variable Importance of XGBoost Classifier

Variable	Importance Score
Rodentia	0.250
Chiroptera	0.175
Primates	0.100
Carnivora	0.085
Pilosa	0.085
Virus_Group	0.062
Diprotodontia	0.060
Eulipotyphla	0.052
Aves	0.050
Artiodactyla	0.027

Table 10.1: Variable Importance Scores — XGBoost Classifier

Rodentia was the most influential predictor (0.250), as rodents are major ecological reservoirs for zoonotic pathogens. **Chiroptera** ranked second (0.175), reflecting bats' role as natural reservoirs for emerging viruses. **Primates** ranked third (0.100) due to phylogenetic proximity to humans. Mammalian reservoir ecology constitutes the dominant biological signal for predicting viral human infectivity.

10.1.10 ROC Curve Analysis

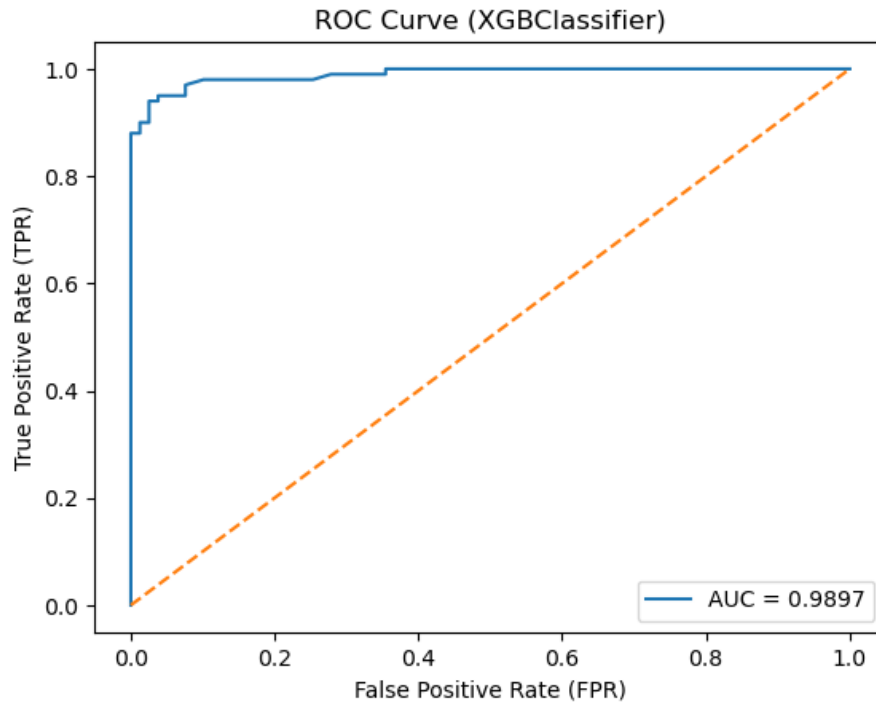


Figure 10.2: ROC Curve — XGBoost Classifier

The ROC curve plots True Positive Rate against False Positive Rate:

$$TPR = \frac{TP}{TP + FN}, \quad FPR = \frac{FP}{FP + TN}$$

The XGBoost model achieved $AUC = 0.99$, surpassing Random Forest ($AUC = 0.9865$). The curve rises sharply toward the upper-left corner, confirming high sensitivity and specificity.

10.1.11 Train vs Test MSE and RMSE

MSE (Brier Score) and RMSE were computed:

$$MSE = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{p}_i)^2, \quad RMSE = \sqrt{MSE}$$

Metric	Train	Test
MSE	0.0289	0.0381
RMSE	0.1701	0.1952

Table 10.2: Train vs Test MSE and RMSE — XGBoost Classifier

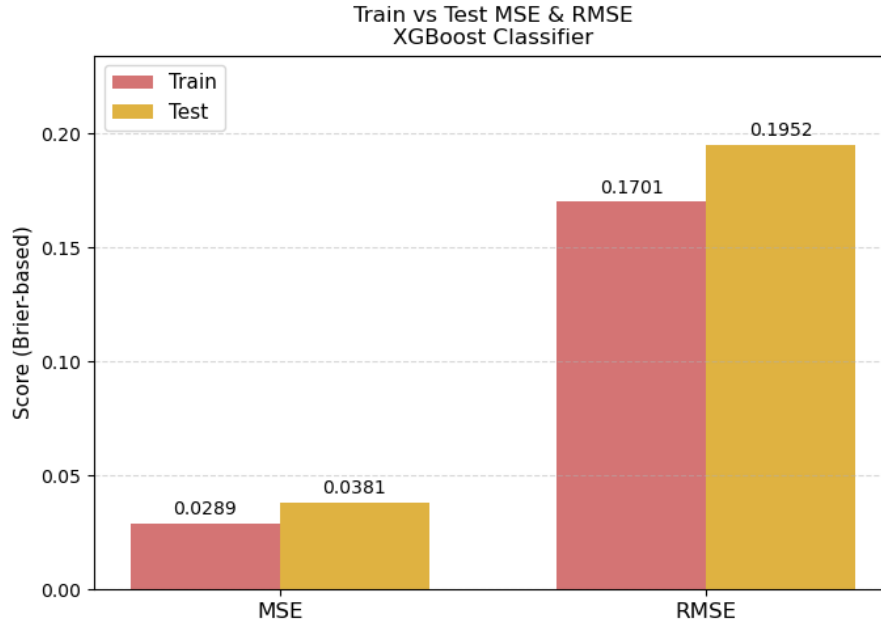


Figure 10.3: Train vs Test MSE and RMSE — XGBoost Classifier

The small gap between Train MSE (0.0289) and Test MSE (0.0381) confirms no severe overfitting. Regularization parameters γ and λ effectively constrain model complexity.

10.1.12 Conclusion

XGBoost achieved the highest predictive accuracy (94.97%) and AUC (0.99) among all models evaluated. The dominance of mammalian reservoir variables in variable importance, combined with strong cross-validation stability, confirms XGBoost as the most reliable framework for predicting viral zoonotic spillover and human infectivity in ecological epidemiological datasets.

Chapter 11

Model Comparison

11.0.1 Comparison Between Random Forest, XGBoost, and Support Vector Machine Classifiers

The predictive performances of the Random Forest, XGBoost, and Support Vector Machine (Linear kernel) classifiers were evaluated using multiple classification metrics, including Accuracy, Area Under the ROC Curve (AUC), Precision, Recall, F1-score, and confusion matrix components. The comparison results are summarized in Table 11.1.

Overall Predictive Performance

The XGBoost classifier achieved the highest overall predictive performance with an accuracy of 94.97% and an AUC score of 0.99, indicating excellent discrimination capability between human-infecting and non-human-infecting viruses. Random Forest also demonstrated strong performance with an accuracy of 93.85% and an AUC score of 0.98.

The Support Vector Machine with Linear kernel achieved a test accuracy of 82.68% and an AUC score of 0.9241. The Linear kernel was selected for comparison because it achieved the highest AUC among the three kernel functions evaluated (RBF: 0.9160, Linear: 0.9241, Polynomial: 0.9016), indicating superior probabilistic separation between human-infecting and non-human-infecting viruses.

Class-Specific Performance

For Class 1 (human-infecting viruses), XGBoost achieved higher recall (0.94) and F1-score (0.95) compared to Random Forest, indicating superior ability to correctly identify zoonotic viruses. The Linear SVM, while achieving very high precision for Class 1 (0.97), exhibited comparatively lower recall (0.71), reflecting the inherent limitation of linear decision boundaries in capturing complex non-linear ecological interaction patterns.

For Class 0 (non-human-infecting viruses), all three models performed well, with Random Forest and XGBoost achieving identical recall (0.96) and the Linear SVM achieving the highest recall (0.97). However, the Linear SVM's lower precision for Class 0 (0.73) indicates a higher rate of false alarms compared to the ensemble methods.

Epidemiological Significance

From an epidemiological surveillance perspective, XGBoost produced the fewest false negatives (6) compared to Random Forest (8) and Linear SVM (29). Minimizing false

negatives is critically important in zoonotic surveillance because missed detections of human-infecting viruses carry substantial public health risk and may delay outbreak response. XGBoost also produced the highest number of true positives (94), correctly identifying more zoonotic viruses than Random Forest (92) and Linear SVM (71).

The Linear SVM produced the fewest false positives (2), meaning it had a lower false alarm rate. However, the high number of false negatives (29) makes it unsuitable as a primary screening tool for emerging zoonotic threats, where sensitivity (recall) is prioritized over specificity.

Macro and Weighted Averages

Macro-average and weighted-average metrics consistently favored XGBoost, followed by Random Forest, with the Linear SVM demonstrating comparatively lower overall balanced performance. XGBoost achieved macro-average precision, recall, and F1-score of 0.95, indicating balanced performance across both classes. Random Forest followed closely with 0.94, while Linear SVM lagged behind with 0.83–0.85.

Conclusion

XGBoost was identified as the best-performing classifier for predicting human infectivity and zoonotic spillover potential in the present study, while Random Forest provides a strong and interpretable alternative. The Linear SVM, despite its lower overall accuracy, confirms the biological separability of the variable space through linear boundaries and provides valuable theoretical complementarity to the ensemble-based approaches.

The superior performance of XGBoost can be attributed to its sequential boosting framework, which iteratively corrects prediction errors, and its regularization parameters (γ , λ), which effectively control overfitting while capturing complex non-linear ecological interactions. The dominance of mammalian reservoir variables (particularly *Rodentia*, *Chiroptera*, and *Primates*) in variable importance analysis across both tree-based models further confirms the central role of host ecology in zoonotic spillover dynamics.

Table 11.1: Performance Comparison Between Random Forest, XGBoost, and SVM (Linear Kernel)

Performance Measure	Random Forest	XGBoost	SVM (Linear)
Accuracy	93.85%	94.97%	82.68%
AUC	0.9865	0.99	0.9241
Precision (Class 0)	0.90	0.93	0.73
Recall (Class 0)	0.96	0.96	0.97
F1-score (Class 0)	0.93	0.94	0.83
Precision (Class 1)	0.97	0.97	0.97
Recall (Class 1)	0.92	0.94	0.71
F1-score (Class 1)	0.94	0.95	0.82
Macro Avg Precision	0.94	0.95	0.85
Macro Avg Recall	0.94	0.95	0.84
Macro Avg F1	0.94	0.95	0.83
Weighted Precision	0.94	0.95	0.86
Weighted Recall	0.94	0.95	0.83
Weighted F1	0.94	0.95	0.83
True Negatives	76	76	77
False Positives	3	3	2
False Negatives	8	6	29
True Positives	92	94	71

Chapter 12

Conclusion

12.1 Summary of Findings

The key findings of this study are summarized below:

- Exploratory analysis confirmed statistically significant associations between human infectivity and virus group ($\chi^2 = 185.92$, $p < 0.001$), continental distribution ($\chi^2 = 21.65$, $p < 0.001$), cross-host diversity ($U = 3082.0$, $p < 0.001$), and total vector diversity ($U = 34646.5$, $p < 0.001$).
- LASSO and Ridge regression consistently identified **Virus_Group**, **Primates**, **Chiroptera**, **Rodentia**, and **Carnivora** as the most influential ecological predictors of human infectivity. LASSO additionally eliminated **Lagomorpha** and **climate_group** as non-informative, producing a sparse and interpretable variable set.
- Random Forest and XGBoost both demonstrated excellent predictive performance with high accuracy and AUC scores. XGBoost achieved the highest classification accuracy (94.97%) and AUC score (0.99), indicating superior discrimination capability between human-infecting and non-human-infecting viruses.
- XGBoost also produced the highest F1-scores and macro-average performance measures, reflecting strong overall classification stability and balanced prediction performance.
- XGBoost produced fewer false negatives compared to Random Forest (6 vs. 8), improving the detection of potentially zoonotic viruses — a critical requirement in epidemiological surveillance where missed detections carry substantial public health costs.
- The superior performance of XGBoost is likely due to its gradient boosting framework, regularization mechanisms (γ and λ), and ability to model complex nonlinear interactions among ecological, virological, and vector-related variables.
- Therefore, **XGBoost was selected as the final predictive model** for zoonotic spillover and human infectivity prediction.

12.2 Limitations

Despite the strong predictive performance achieved, several limitations must be acknowledged:

- **Surveillance Bias:** The dataset was compiled from published epidemiological records, which inherently over-represent well-studied viruses from regions with strong surveillance infrastructure. Viruses from under-surveilled tropical and rural regions may be systematically absent, potentially biasing model predictions toward well-characterized pathogens.
- **Binary Outcome Simplification:** The target variable `homo_infected` is binary, collapsing the full spectrum of human infection severity — from mild febrile illness to fatal hemorrhagic fever — into a single category. This simplification may obscure important differences in clinical and epidemiological significance.
- **Sparse Host Variables:** Several reservoir host variables contained over 95% zero values, limiting their independent predictive contribution and potentially introducing instability in coefficient estimation for regularized regression models.
- **Static Ecological Data:** The dataset represents a cross-sectional snapshot of virus-host-vector associations. Dynamic ecological changes driven by climate change, deforestation, urbanization, and land-use modification are not captured, limiting the model’s ability to project future spillover risk under changing environmental conditions.
- **Absence of Genomic Features:** The present framework relies entirely on ecological and taxonomic predictors. Incorporating viral genomic sequence features — such as genome length, GC content, codon usage bias, and mutation rate proxies — could substantially improve classification accuracy and biological interpretability.
- **Correlation vs. Causation:** While machine learning models identify strong predictive associations, they do not establish causal relationships. The identified predictors may be correlates rather than direct drivers of zoonotic emergence, and causal inference methods would be required to distinguish these.

12.3 Future Work

Several directions remain open for future investigation:

- **Genomic Integration:** Combining ecological predictors with viral genomic features — such as genome length, GC content, codon usage bias, and mutation rate proxies — could improve both predictive accuracy and mechanistic understanding of zoonotic emergence. Recent advances in deep learning for genomic sequence analysis offer promising avenues for this integration.
- **Longitudinal Modeling:** Extending the framework to incorporate temporal surveillance data would allow modeling of dynamic changes in zoonotic risk over time and geographic space. This would enable early warning systems that track emerging risk patterns in near real-time.

- **Probabilistic Risk Scoring:** Developing a continuous zoonotic spillover risk score — rather than a binary classifier — would provide finer epidemiological resolution for prioritizing surveillance efforts and allocating public health resources more efficiently.
- **Causal Inference Framework:** Future work could apply causal inference methods (e.g., structural equation modeling, directed acyclic graphs, or instrumental variables) to distinguish true ecological drivers of human infectivity from spurious correlational associations, improving the scientific validity of predictive claims.
- **Expansion to Novel Virus Discovery:** The trained XGBoost model could be applied prospectively to newly discovered viruses, providing rapid preliminary assessment of their human infection potential before laboratory validation becomes available. This would accelerate risk assessment during early outbreak investigation.
- **Integration with Climate and Land-Use Data:** Incorporating high-resolution climate projections and land-use change data could enhance the model’s ability to forecast future spillover hotspots under scenarios of environmental change.

12.4 Concluding Remarks

This study demonstrates that ecological host breadth, wildlife reservoir diversity, viral genomic classification, and vector taxonomy collectively provide strong and interpretable signals for predicting zoonotic virus infectivity in humans. By integrating regularization-based variable selection with ensemble machine learning, the framework advances the transition from reactive outbreak surveillance toward proactive predictive epidemiology.

The consistent dominance of mammalian reservoir variables — particularly *Rodentia*, *Chiroptera*, and *Primates* — across all modeling approaches reinforces the central role of host ecology in zoonotic spillover dynamics. The superior performance of XGBoost, with its ability to capture complex nonlinear interactions and its regularization mechanisms that prevent overfitting, establishes it as a robust and reliable tool for epidemiological risk stratification.

The findings have direct implications for public health policy and surveillance strategy. By identifying high-risk viral pathogens before they cause large-scale outbreaks, predictive frameworks can guide targeted surveillance, inform vaccine development priorities, and support evidence-based resource allocation for emerging infectious disease preparedness.

Ultimately, this study contributes a reproducible, biologically meaningful, and computationally efficient framework for predicting zoonotic spillover potential — bridging the gap between ecological theory and practical epidemiological application in an era of accelerating infectious disease emergence.

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Chapter 13

Acknowledgment

Acknowledgment

I would like to express my sincere gratitude to my supervisors, **Prof. Bhawasti Ganguly**, **Prof. Gaurangadev Chattopadhyayee**, and **Prof. Sougata Sen Roy**, Department of Statistics, University of Calcutta, for their invaluable guidance, constant encouragement, and insightful suggestions throughout the course of this project. Their expertise and constructive feedback have been instrumental in shaping both the analytical framework and the overall direction of this work.

I am also deeply thankful to **Prof. Asis Kumar Chattopadhyayee** for his generous guidance and support in the areas of machine learning and deep learning, which greatly strengthened the methodological foundation of this study.

I am deeply grateful to the faculty members of the Department of Statistics, University of Calcutta, for their dedicated teaching and for providing a stimulating academic environment that has greatly enriched my understanding of statistical methods and their applications.

I would also like to acknowledge the Mendeley Data Repository, along with the World Health Organization (WHO), the National Center for Biotechnology Information (NCBI), GenBank, and the International Committee on Taxonomy of Viruses (ICTV), for making the virological surveillance data publicly available, without which this study would not have been possible.

Finally, I wish to thank my family and friends for their unwavering support and encouragement throughout my academic journey.

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July 2026